When Pedagogical Alignment Is Not Enough

Cognitive Cost and Expertise-Conditional Outcomes in an RCT of a Scaffolded Sales Game

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Abstract

Background: Game-based learning (GBL) is often justified through the assumption that well-aligned game mechanics can make practice more engaging while supporting learning. Alignment between learning mechanics and game mechanics does not by itself show that learners can absorb the added mechanics without cognitive cost. This thesis tests that assumption in a randomized evaluation of a scaffolded sales game.

Methods: A randomised controlled trial ($N = 63$ completers) compared a scaffolded digital sales game with a content-matched conventional digital module. Sales self-efficacy (SE) and performance, operationalized through ADAPTS-SV adaptive selling beliefs, were analysed using ANCOVA with pre-test scores as covariates. Cognitive load, engagement, and transfer intention were examined using Welch t-tests. Exploratory follow-up analyses examined in-game behavioral telemetry, mediation within the game group, and expertise-conditional patterns.

Results: The game did not outperform the control on the primary outcomes. ANCOVA showed a marginal trend favouring the control on SE ($F(1{,}60) = 3.56$, $p = .064$, $\eta_p^2 = .056$) and no significant difference on performance ($F(1{,}60) = 2.22$, $p = .141$). The clearest between-group result was that the game produced substantially higher perceived cognitive load ($d = 0.71$, $p = .006$) without corresponding aggregate gains in engagement, transfer intention, SE, or performance. Exploratory readiness interactions showed that outcomes depended on pre-intervention readiness (pre-performance × condition: $F(1{,}59) = 11.65$, $p = .0012$; pre-SE × condition: $F(1{,}59) = 5.52$, $p = .022$). Baseline cluster translation illustrated the same pattern: novice learners showed a large control advantage on performance gain, while intermediate and advanced learners were closer to parity. Within the game group, SE fully mediated the effect of engagement on performance (indirect effect = 1.10, 95% CI [.43, 2.24]), suggesting a viable motivational pathway among learners who could absorb the game.

Conclusions: The study is best read as a theoretically informative disconfirmation: pedagogical alignment was present and still left additional cognitive cost without superior immediate outcomes. The findings motivate a proposed *Cognitive Budget Framework* for extending LM–GM reasoning, while stopping short of validating it. Within this framework, *mechanics acquisition cost* refers to the working-memory demand created when learners must learn how to operate game mechanics before those mechanics can carry pedagogy. The thesis positions this framework as a worked example of why game-based training should budget mechanics cost against learner readiness.

Keywords: digital game-based learning, cognitive load, randomised controlled trial, Cognitive Budget Framework, mechanics acquisition cost, sales self-efficacy, adaptive selling

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Chapter 1

Introduction

1.1 Background

Digital game based learning (DGBL) has been associated with better learning performance, improved learning attitudes, and higher motivation than traditional face-to-face instruction (Chang et al., 2020) and online virtual learning environments (Hernández and Moreno, 2019). It has been used to support critical thinking and argumentation (Noroozi et al., 2020), leadership skills (Sousa and Costa, 2014), and corporate competence development (Senderek et al., 2015). Its uptake in workplace training remains uneven. Tay et al. (2022) identify persistent barriers, including limited available solutions, lack of authoring expertise, doubts about efficacy, and negative associations with games in some professional settings (Bösche and Kattner, 2011). These barriers make design quality central. DGBL needs a clear pedagogical structure so that game elements support learning goals instead of adding avoidable activity. The learning mechanics–game mechanics framework (LM–GM) offers one such structure by mapping game mechanics to intended learning mechanics (Arnab et al., 2015).

Corporate sales training is a fitting test case because sales education requires practice, feedback, and confidence under realistic constraints (Cummins et al., 2013; Tay et al., 2022). Existing DGBL studies still struggle to explain why a game succeeds or fails, especially when research designs lack rigorous comparison conditions (All et al., 2021). Many studies cite engagement as the driver of improved learning outcomes (Dahalan et al., 2024; Woo, 2014), but engagement does not identify the mechanism. Cognitive load theory (Sweller et al., 2011) and self-efficacy theory (Bandura, 1977, 1986) offer a more precise way to examine whether a sales game becomes learnable, confidence-building, and effective (Barling and Beattie, 1983; Hung et al., 2014).

The DGBL evidence base also has methodological limits. Randomized controlled trials remain scarce, which constrains comparison with conventional designs and weakens causal claims about learning outcomes (All et al., 2021; Hainey et al., 2016; Tay et al., 2022). All et al.'s (2021) review identifies recurring issues that shape this study: weak random assignment in online recruitment, limited attention to instructional time as an attrition factor, and limited measurement of motivation to start playing. In this trial, the in-game behavioral telemetry shows that many game dropouts occurred at phase transitions with high time demands. Entry motivation also appears as selective attrition, giving the dropout pattern interpretive value as part of the intervention record. These design choices and their consequences are discussed further in Section 4 and Section 5.2. The sales domain also needs direct comparisons between pedagogically grounded DGBL and standard digital formats. Scaffolding strategies in sales games may influence cognitive load, self-efficacy, adaptive selling, and transfer (Chang et al., 2018; Evans et al., 2024; Cardinali et al., 2025), yet those mechanisms remain under-tested. Digital card games are a plausible format for this problem because they can segment decisions and support repeated practice (Kordaki and Gousiou, 2017).

1.2 Research Objective

To guide and focus the investigation, the following research questions are formulated:

- **Research question 1 (RQ1):** How does a scaffolded game-based learning (DGBL) intervention compare to a content-matched conventional digital training module in terms of cognitive load, sales self-efficacy, performance, engagement, and transfer intention?

This primary research question compares the two training conditions across the main psychological and applied sales outcomes. The design expectation was that the scaffolded game would produce lower cognitive load and stronger motivational and performance outcomes than the conventional module. The analysis treats the disconfirmation of that expectation as part of the theoretical result.

- **Research question 2 (RQ2):** Within and around the scaffolded DGBL sales training, how do perceived cognitive load, learner engagement, sales self-efficacy, in-game behavioral telemetry, and pre-intervention readiness help explain the observed treatment pattern?

This second research question focuses on mechanism and boundary conditions. It investigates whether cognitive load, engagement, and self-efficacy behaved as the original design logic expected; whether in-game behavioral telemetry supports the interpretation that the game created additional operating demands; and whether pre-intervention readiness explains why the same aligned game mechanics produced different value for different learners.

1.3 Significance of the Study

This study contributes to educational technology, instructional design, corporate training, and sales education by testing a pedagogically aligned DGBL intervention against a content-matched digital module.

1.3.1 Theoretical Contributions

The study stress-tests a DGBL intervention designed around cognitive load theory (CLT), self-efficacy theory, motivation theory, and LM–GM alignment. The aggregate result — higher cognitive load in the game condition without corresponding gains in self-efficacy or performance — documents a case in which pedagogical alignment produced additional cognitive overhead. The thesis contributes the *Cognitive Budget Framework* as a proposed extension to LM–GM reasoning. Its operational construct is *mechanics acquisition cost*: the working-memory demand required to operate game mechanics to the point where they can support the intended pedagogy. The data motivate this framework as a worked example; they do not validate it.

1.3.2 Practical Contributions

The study evaluates a scaffolded, card-based digital game for sales training alongside a content-matched conventional module. The aggregate null-to-marginal result has practical value: a well-designed, LM–GM-grounded game did not automatically outperform a conventional digital module for a mixed-readiness population. The exploratory expertise-conditional pattern suggests that deployment should account for learner prior knowledge, with novice learners receiving direct instruction or lighter onboarding before the full game. The higher attrition rate in the game condition (41.8%) also matters for deployment. Organizations should treat completion as part of the intervention outcome, not just as a statistical filter.

1.3.3 Methodological Contributions

The study also contributes methodologically. It demonstrates a randomised online evaluation of a DGBL intervention in a professional training context, pairing survey-based psychometric outcomes with in-game behavioral telemetry — a combination that remains comparatively rare in the GBL literature (All et al., 2021). The clustering approach was restricted to pre-intervention features to avoid defining groups with the same post-test or gain variables used to evaluate them. Baseline readiness interaction models provide the stronger evidence for learner-readiness effects, while the cluster solution translates that result for interpretation. Two intention-to-treat sensitivity analyses bound plausible treatment effects under different attrition assumptions. Finally, the study highlights a measurement limit: brief global cognitive-load scales can detect cognitive cost, but they cannot adjudicate cleanly among intrinsic, extraneous, and germane load.

Together, these contributions support a more selective design and deployment logic for corporate sales games.

1.4 Delimitations of the Study

This study's findings should be interpreted within the following scope:

- **Participant Generalizability:** While targeting novice sales professionals, open online recruitment means the sample may be diverse; results primarily apply to this self-selected group.
- **Intervention Specificity:** The study evaluates a specific scaffolded card-based DGBL ("Improve Your Sales Game"); findings may not generalize to all DGBL formats or all sales training contexts.
- **Outcome Scope:** The focus is on self-reported adaptive selling, transfer-oriented perceptions, and related immediate post-training outcomes, not the full spectrum of sales competencies or actual on-the-job performance.

- **Measurement Timeline:** Outcomes are assessed immediately post-intervention, without long-term follow-up on retention or real-world skill transfer.

Chapter 2

Literature Review

2.1 Sales Education and the Case for DGBL

2.1.1 Why DGBL Fits Sales Education

Sales roles are performance-oriented and skills-based, and the competencies they demand shift with market conditions, which makes ongoing development a structural requirement (Cummins et al., 2013). Sales education needs instructional formats that can deliver repeated, safe, and scalable practice under realistic organisational constraints. DGBL is examined here because it can combine scenario practice, feedback, learner agency, and traceable digital interaction in a single training environment.

Engagement, Motivation, and Active Learning

Traditional training methods can struggle to sustain the engagement and motivation on which skill development depends (Woo, 2014). DGBL supports active participation and intrinsic motivation through interactive mechanics, narrative, challenge, and immediate feedback (Tay et al., 2022; Chang et al., 2018). Engagement is often proposed as the driver of improved learning outcomes, including in corporate training (Hainey et al., 2016). In this thesis, engagement becomes relevant when it connects to manageable cognitive effort, mastery experiences, and transferable adaptive selling beliefs.

Safe and Realistic Practice for Complex Skill Development

Sales success often hinges on interpersonal and strategic skills such as negotiation, objection handling, and adapting communication to client needs (Bussière, 2017). DGBL can simulate realistic sales scenarios and give trainees a safe environment for practice without real-world consequences (Senderek et al., 2015; Dahalan et al., 2024). This sandboxed format supports experimentation, learning from mistakes, and iterative skill refinement (Chen et al., 2018).

This need for practice is especially visible in adaptive selling. Learners must read customer cues, diagnose a selling situation, select an appropriate response, and revise behaviour when the situation changes. Passive exposure to sales concepts cannot fully exercise that sequence. Repeated application under ambiguous customer-like conditions is needed, especially for novices who still need protection from the risk and social cost of real failure.

Problem-Solving, Critical Thinking, and Learner Agency

Effective sales professionals are adept problem-solvers who can critically assess situations and devise appropriate strategies. DGBL can be designed to present learners with ill-defined problems or multifaceted challenges that require analytical thinking, strategic decision-making, and creative solutions (Noroozi et al., 2020). Such designs can also enhance learner agency by allowing individuals to make meaningful choices and observe the consequences of those choices within the game environment.

Scalability, Consistency, and Corporate Upskilling

Organizations often face substantial training demands as they seek to ensure that sales teams possess current knowledge and skills in response to new products, evolving market conditions, and the general need for continuous professional upskilling (Hernández and Moreno, 2019). DGBL, as a digital modality, offers a scalable and consistent way to deliver high-quality training content across geographically dispersed teams, thereby supporting efficient competency development (Tay et al., 2022).

DGBL fits sales education when its affordances support specific training needs: repeatable practice, immediate feedback, consequence-bearing decisions, and scalable delivery. The design problem is also cognitive. The game must make complex sales situations easier to process while preserving the motivational benefits of interactive practice. This is especially important for adult professional learners. Tay et al. (2022) note that professional-upskilling contexts involve varied prior knowledge and work experience, and that adults do not benefit from every possible game element. Game features need a pedagogical reason.

Meta-analytic reviews offer qualified support for DGBL. Wouters et al. (2013) found that serious games outperformed conventional instruction on learning outcomes ($d = 0.29$) and retention ($d = 0.36$), with stronger effects when games included multiple practice sessions, collaboration, and supplementary instruction. Merchant et al. (2014) found an overall effect of $d = 0.33$ for simulation-based learning. Both reviews draw heavily on student populations in formal educational settings. Their aggregate effects cannot be transferred directly to professional adult sales training, where learner agency, prior domain expertise, and motivation differ. All et al.'s (2021) review gives more attention to workplace and professional DGBL contexts and offers a more cautious picture: games need design quality, instructional alignment, and rigorous comparison conditions before claims of superiority are warranted.

All et al.'s critique shifts the evaluation question from general engagement to mechanism. Their review calls for stronger empirical comparison, clearer attention to the intervention and comparison condition, and better evidence about how game features produce outcomes. Their corporate-context feasibility evidence is directly relevant: in one corporate comparison, a DGBL intervention and an instructional video both improved learning relative to business-as-usual exposure, while the game added no motivational or learning value over the simpler digital alternative. The distinction between absolute effectiveness and relative effectiveness matters here. This thesis asks whether the game adds value beyond a content-matched conventional module. The study responds by evaluating a content-matched sales game and control module, measuring psychological outcomes, and using behavioral telemetry to interpret how learners moved through the game. The design assumed that pedagogical alignment would lower unnecessary cognitive load: clean LM–GM mapping should let interactivity segment and scaffold sales reasoning.

This position is consistent with the maturation of gameful-design research. Nacke and Deterding (2017) argue that the field has moved from broad effectiveness questions toward more differentiated questions of how, when, and for whom particular designs work. That shift matters for DGBL because a serious game is a configuration of mechanics, instructional content, learner interpretation, and context. This thesis treats the sales game as a mechanism-bearing intervention, not as a generic "game" expected to outperform conventional instruction by being more engaging.

2.1.2 Current Approaches to Sales Education

Traditional and Experiential Methods

Sales education has moved beyond product training and now aims to develop the competencies required in increasingly complex market environments. Pedagogical strategies need to connect theoretical knowledge with practical application so that training can transfer to organizational performance (Tho and Trang, 2015).

Traditional methods such as lectures, assigned readings, and classroom discussions still deliver foundational knowledge, including sales theories, ethical guidelines, and market dynamics. Sales also requires experiential learning. Cummins et al. (2013) note the widespread use of role-play and simulation in sales pedagogy. These activities let learners practice sales interactions, from initial approaches to closing and objection handling, in a controlled environment that can build skill and self-efficacy (Knight et al., 2014).

Case Study Analysis

Analyzing real-world or hypothetical business cases helps learners develop diagnostic and strategic thinking skills that are directly relevant to sales challenges.

Field Projects and Internships

Direct engagement in sales activities, whether through internships or structured "selling lab" projects, provides hands-on experience and supports understanding of the sales process through application (Bussière, 2017).

The emphasis on experiential methods reflects the view that active participation and "learning by doing" support skill acquisition and retention in professional disciplines. These approaches foster procedural skills, critical thinking, problem-solving, and interpersonal communication.

These methods still face limits. Transfer from training environments to on-the-job performance can be inconsistent and is shaped by the training design and the organizational context (Tho and Trang, 2015). Resource-intensive methods such as role-play or mentored field experience are also difficult to scale consistently. Assessing the skills developed through such methods often requires detailed rubrics and remains partly subjective.

The sales profession itself is also changing. The traditional linear "seven steps of selling" model (Moncrief and Marshall, 2005) remains useful, but it is increasingly supplemented or recontextualized by more dynamic, customer-centric, and digitally mediated approaches (Cardinali et al., 2025). Sales education must prepare learners for complexity, adaptability, and value co-creation.

DGBL can translate several principles of experiential learning into an interactive digital format. It also offers scalability, consistency, immediate feedback, adaptive pathways, and the capture of performance data, which address some limits of traditional experiential methods.

2.2 Theoretical and Design Foundations

2.2.1 LM–GM Framework

Sales DGBL needs a design framework that links game activity to learning purpose. Arnab et al. (2015) formulated the learning mechanics–game mechanics framework (LM–GM) by identifying learning mechanics categorized by instructional purpose. The model shows how game mechanics can be linked to learning mechanics across increasing levels of cognitive complexity.

Correspondingly, the framework outlines numerous game mechanics that can be employed to realize these learning mechanics. Examples include:

- **Guidance & Feedback:** Tutorials, hints, pop-up information, and direct feedback messages such as success/failure indicators or explanatory notes.
- **Participation & Action:** Quests, tasks, levels, puzzles, resource management challenges, and the use of tokens or point systems.
- **Exploration & Discovery:** Open environments, branching narratives, hidden information elements, and interactive objects.
- **Reflection & Collaboration:** In-game dialogue systems, forums, team-based challenges, or voting mechanisms.
- **Motivation & Ownership:** Customizable avatars, leaderboards, collection mechanics, and clear progression systems.

The LM–GM framework encourages a design process that begins with learning outcomes,

GAME MECHANICS	THINKING SKILLS	LEARNING MECHANICS	LOTS to HOTS
○ Design/Editing ○ Infinite Game play ○ Ownership ○ Protégé Effect ○ Status ○ Strategy/Planning ○ Tiles/Grids	CREATING	○ Accountability ○ Ownership ○ Planning ○ Responsibility	
○ Action Points ○ Assessment ○ Collaboration ○ Communal Discovery ○ Resource Management ○ Game Turns ○ Pareto Optimal ○ Rewards/Penalties ○ Urgent Optimism	EVALUATING	○ Assessment ○ Collaboration ○ Hypothesis ○ Incentive ○ Motivation ○ Reflect/Discuss	
○ Feedback ○ Meta-game ○ Realism	ANALYSING	○ Analyse ○ Experimentation ○ Feedback ○ Identify ○ Observation ○ Shadowing	
○ Capture/Elimination ○ Competition ○ Cooperation ○ Movement ○ Progression ○ Selecting/Collecting ○ Simulate/Response ○ Time Pressure	APPLYING	○ Action/Task ○ Competition ○ Cooperation ○ Demonstration ○ Imitation ○ Simulation	
○ Appointment ○ Cascading Information ○ Questions And Answers ○ Role-play ○ Tutorial	UNDERSTANDING	○ Objectify ○ Participation ○ Question And Answers ○ Tutorial	
○ Cut scenes/Story ○ Tokens ○ Virality ○ Behavioural Momentum ○ Pavlovian Interactions ○ Goods/Information	RETENTION	○ Discover ○ Explore ○ Generalisation ○ Guidance ○ Instruction ○ Repetition	

Figure 2.1: LM–GM model classified based on Bloom's ordered thinking skills, adapted from Arnab et al. (2015).

then identifies the learning mechanics needed to achieve them, and then selects game mechanics to implement those learning mechanics. This keeps the game's design anchored to pedagogical purpose.

For this thesis, LM–GM alignment also carries a cognitive assumption. Mechanics chosen to enact a learning mechanic should reduce search, clarify attention, and make practice more structured. Pedagogical alignment is expected to help manage cognitive load: cards, tokens, progression, and feedback should make the sales task easier to organize in working memory than an unstructured or decorative game layer. This is the design logic tested in the trial.

That logic has limits. Arnab et al. (2015) describe LM–GM as a descriptive mapping tool, not a guarantee of effectiveness. They also note that the framework does not yet include a player model, even though adaptation and personalization matter in serious game design. This limitation matters here: LM–GM can show that a mechanic is intended to serve a learning function, but it cannot show that the target learner has enough readiness to absorb that mechanic while learning the domain content.

For instance, if a learning objective is for sales trainees to classify different types of customer objections, which involves learning mechanics such as identification and discrimination, the game might employ card-sorting challenges or scenario-based quizzes in which players must correctly categorize presented objections to progress. For objectives involving negotiation skills, which rely on experimentation and reflection, the game might use branching dialogue systems with varied outcomes and provide after-action reviews as a feedback mechanic to encourage reflection on choices and consequences.

The LM–GM framework also supports analysis of existing serious games. It lets researchers and practitioners deconstruct how specific game features contribute, or fail to contribute, to learning. In their application of the framework to *Re-Mission*, Arnab et al. (2015) note that even LM–GM-designed games may include "extraneous (pedagogy-independent) game mechanics" that make learning occur "only tangentially." This analytic capacity is one reason the framework is used here, and it also shows why empirical testing is needed: design-level alignment must still translate into cognitive-level learning gain.

2.2.2 Foundations of Game-Based Learning

Plass, Homer and Kinzer (2015) provide a broader theoretical anchor for game-based learning than design-mapping frameworks alone. They describe game-based learning as resting on four interlocking foundations: cognitive, motivational, affective, and sociocultural. This perspective clarifies why a well-aligned game can still produce uneven outcomes across learners. In this study, LM–GM explains how mechanics were mapped to intended learning functions; Plass et al. help locate the cognitive, motivational, and affective consequences of that design.

This thesis contributes most directly to the *cognitive* foundation in Plass et al.'s model. Game mechanics need to be pedagogically aligned and cognitively absorbable for the target learner population. The motivational and affective foundations remain visible through engagement and self-efficacy, which are treated as downstream mechanisms whose expression depends partly on whether learners can absorb the mechanics inventory without exhausting working-memory capacity. The Cognitive Budget Framework is positioned as an operational proposal nested inside Plass's cognitive foundation and focused on the mechanics acquisition cost that LM–GM-aligned designs may still impose.

Plass et al.'s framework also clarifies why engagement cannot be treated as a single undifferentiated benefit. Games can support behavioral, affective, cognitive, and sociocultural engagement, but learning depends on whether those forms of engagement lead back to cognitive engagement with the learning mechanic. They also emphasize adaptation and personalization, including the need to measure variables such as prior knowledge or self-regulation before a game can respond appropriately. This anticipates the readiness issue examined here.

2.2.3 Gameful Learning, Mechanism, and Learner Context

Research on gamification and gameful learning offers a caution for interpreting DGBL effectiveness. Gamification and DGBL differ, and both assume that game elements can alter how learners engage with instructional tasks. The stronger claim is that specific elements affect specific psychological or behavioral processes that contribute to learning. For this study, mechanics alignment must be evaluated through mediators and boundary conditions as well as aggregate outcome differences.

Empirical work in this area shows why the mechanism needs testing. Mekler et al. (2017) found that points, levels, and leaderboards increased output quantity in an image-annotation task, yet did not improve output quality, perceived competence, autonomy, or intrinsic motivation. Their hypothesized competence-mediated motivation pathway was not supported. For this thesis, the lesson is that game mechanics can change observable behavior without activating the psychological pathway designers expect. Engagement and telemetry need to be examined alongside self-efficacy, transfer intention, adaptive selling, and cognitive load.

Learner context also matters. Applying the Technology-Enhanced Training Effectiveness

Model to gamification, Landers and Armstrong (2017) found that expected value for gamified instruction depended on prior game attitudes and video-game experience. Participants with positive attitudes and higher experience anticipated more benefit from gamified instruction; those with lower attitudes and experience anticipated more benefit from traditional instruction. Their explanation connects directly to cognitive load: when learners have limited experience with a training technology, part of their effort is spent learning how to use the technology instead of learning the instructional content. Their moderator was game-related readiness and their outcome was anticipated training value, so the evidence does not prove sales-expertise moderation. It does establish a broader principle: gameful training is readiness-sensitive. In this thesis, readiness is operationalized through baseline sales self-efficacy and baseline ADAPTS-SV. The parallel question is whether the learner enters with enough prior schema and confidence for the mechanics layer to become useful.

Proulx, Romero and Arnab (2017) extend the same concern into LM–GM by interpreting learning and game mechanics through self-determination theory. Their LMGM–SDT perspective argues that the motivational meaning of a mechanic depends on implementation. Progression can feel controlling if it forces a rigid path, or autonomy-supportive if it offers a suggested route while preserving meaningful choice. This helps justify treating self-efficacy and engagement as empirical mechanisms.

2.2.4 Cognitive Load Theory

Cognitive load theory (CLT) supplies the main psychological mechanism behind the intervention design. The thesis treats game effectiveness as a mechanistic design claim. The original design assumption was that a pedagogically aligned game could lower unnecessary cognitive load by structuring attention, segmenting decisions, and providing immediate feedback during complex sales practice. CLT begins from the claim that learning depends on the organization of knowledge in memory. Similar instructional conditions can still lead to different outcomes, which raises a cognitive question: how much working memory is required to learn something at a given time?

Sweller et al. (2011) describe CLT on the basis that working memory is limited in both capacity and duration, whereas long-term memory is virtually unlimited. The crux of learning lies in transferring information from the temporary space of working memory into the more durable structures of long-term memory. This occurs through schema construction and, over time, through automation.

CLT categorizes cognitive load into three types: intrinsic, extraneous, and germane load. Intrinsic load is the inherent difficulty of the material, influenced by element interactivity. Sales training is naturally high in element interactivity because learners must coordinate customer cues, sales stages, objections, response options, and relationship consequences. Extraneous load is created by poor instructional design, distracting materials, or unnecessary embellishment and should be minimized as much as possible. Germane load is the mental effort that directly contributes to learning and schema formation. Instruction should therefore aim to scaffold the intrinsic complexity of sales reasoning, reduce avoidable extraneous processing, and preserve cognitive capacity for reflection, deep processing, and transfer.

These distinctions help educators diagnose learner difficulty, reduce extraneous load by streamlining content, scaffold intrinsic load to match learner readiness, and preserve capacity for germane processing. In DGBL, this logic gives pedagogical alignment its intended mechanism. A card game can segment customer information into discrete units, constrain the response set, cue the relevant sales stage, and provide feedback before the next scenario. Under that design logic, the game should lower the cognitive cost of practicing adaptive sales reasoning. CLT is still open to criticism. It depends on a simplified representation of memory and may underspecify emotional, episodic, or procedural influences on learning. It also risks circularity when failures are retrospectively labelled as extraneous load and successes as germane load. As Popper, discussed in Church (1975), argued more broadly, a theory that can explain everything risks becoming difficult to falsify. These limits matter in sales education, where much knowledge is tacit and only partly formalized.

2.2.5 Expertise Reversal and Learner Readiness

The expertise reversal effect provides the learner-readiness mechanism for this thesis. Kalyuga et al. (2003) argue that the effectiveness of an instructional technique can depend on the learner's prior knowledge because schemas stored in long-term memory reduce working-memory demands. Techniques that support one expertise level may become redundant, inefficient, or detrimental at another level. Kalyuga (2007) frames this as a learner-tailoring problem: instructional systems should adapt support and complexity to what learners already know.

This principle is especially relevant for game-based learning because game mechanics add an operating layer on top of domain content. A novice learner may need to use limited working memory to understand rules, tokens, phase transitions, and interface actions before the sales content can be processed productively. A more experienced learner may already possess enough sales schema to absorb the same mechanics layer with less disruption. In this thesis, expertise reversal is treated as a readiness hypothesis: low-readiness learners are expected to be most vulnerable to mechanics acquisition cost, whereas higher-readiness learners are expected to tolerate or benefit from the game more readily.

This does not require claiming that the sales content differed across conditions. At the design level, the intrinsic content was held constant: both arms covered the same broad sales stages and scenario family. At the learner level, Kalyuga's account makes clear that cognitive load is shaped by the learner processing the material. Prior schema changes how many elements must be held and coordinated in working memory. The Cognitive Budget Framework treats the game-control comparison as content-equivalent with potentially different experienced load.

Recent adult DGBL evidence supports the same readiness-sensitive interpretation. Chang and Yang (2023) found that scaffolding type interacted with adult learners' cognitive style in predicting emotion, cognitive load, and learning performance: hard scaffolding benefited serialist learners, whereas soft scaffolding benefited holist learners. The specific cognitive-style taxonomy is secondary here. The relevant pattern is that DGBL support is conditional on learner characteristics.

2.2.6 Self-Efficacy

DGBL can foster self-efficacy when it manages cognitive load well enough for learners to experience progress. As described by Bandura (1977), self-efficacy beliefs are shaped through four sources of information: mastery experiences, vicarious experiences, verbal persuasion, and physiological or affective states.

DGBL interventions can be designed to use these sources. For mastery experiences, DGBL can provide repeated opportunities for task completion and problem-solving. By segmenting complex sales scenarios into manageable tasks, learners can accumulate a sequence of successes. These achievements build confidence, especially when cognitive load has been managed well enough for tasks to feel achievable.

DGBL can also provide vicarious experiences through models of successful performance, such as non-player characters or simulated peers navigating difficult sales interactions. Verbal persuasion can be built into feedback systems through guidance and reinforcement. Positive affective states may emerge when the game is engaging and appropriately challenging without becoming overwhelming. By reducing overload and learning-related anxiety, and by supporting enjoyment or flow-like experience (Chang et al., 2018), DGBL can help learners associate the learning process with efficacy-building feelings.

This study treats self-efficacy as a test of whether the game environment became psychologically productive. If the mechanics layer produces mastery experiences, self-efficacy should increase and help explain adaptive selling outcomes. If the mechanics layer absorbs working memory before learners experience mastery, self-efficacy may fail to improve even when the design is pedagogically aligned.

2.2.7 Engagement

Alongside cognitive load and self-efficacy, learner engagement is a third construct in the intervention logic. Engagement refers to the intensity and quality of a learner's active involvement in a learning activity (Christenson et al., 2012). It is commonly described through behavioral, emotional or affective, and cognitive dimensions (Fredricks et al., 2004):

- **Behavioral Engagement:** Observable actions such as participation in tasks, time on task, effort, persistence in the face of difficulty, and adherence to learning protocols.
- **Emotional/Affective Engagement:** Feelings and affective responses to the learning activity, including interest, enjoyment, enthusiasm, and a sense of connection to the task.
- **Cognitive Engagement:** Psychological investment in learning, including focused attention, use of learning strategies, self-regulation, and willingness to grapple with complex ideas and exert mental effort.

Research links learner engagement with achievement, skill acquisition, and knowledge retention (Hainey et al., 2016; Hernández and Moreno, 2019; Tay et al., 2022). In workplace training, engagement matters because adult learners often seek training that is relevant, stimulating, and applicable to their roles (Knowles et al., 2020). Disengagement can lead to superficial learning, poor transfer, and a perception of wasted training investment.

DGBL is frequently cited for its potential to enhance learner engagement compared with more passive instructional methods (Dahalan et al., 2024; Liu and Zhou, 2025). Several characteristics of well-designed games contribute to this potential:

- **Intrinsic Motivation:** Games often tap into challenge, curiosity, fantasy, and control.

- **Clear Goals and Rules:** Games provide explicit objectives and rules that can improve focus and behavioral engagement.
- **Immediate Feedback:** Continuous feedback helps learners understand the consequences of their decisions and remain cognitively invested (Chang et al., 2020).
- **Sense of Agency and Control:** Games often let players make meaningful choices that affect progression or outcome.
- **Narrative and Storytelling:** Story-driven contexts can enhance emotional and cognitive engagement (Chen et al., 2018).
- **Challenge and Mastery:** Appropriately balanced challenge can motivate persistence and contribute to both affective and behavioral engagement.

This study examines whether the proposed DGBL intervention fostered higher engagement than the content-matched conventional module. Engagement is also treated mechanistically: if the game produces value, that value should appear through its relationship with self-efficacy, transfer intention, and adaptive selling performance. This follows Mekler et al.'s (2017) caution that observable engagement or behavior change does not by itself identify the motivational mechanism through which the design operates.

2.2.8 Sales Competency

Workplace training aims to improve learners' capabilities and job performance. In sales, this means developing sales competency: an integrated set of knowledge, skills, abilities, and related personal characteristics that enable effective task performance (Titus, 1994).

Sales competency is multifaceted. While specific emphases vary across roles and industries, several core areas recur in the literature (Cummins et al., 2013): product and market knowledge; sales process skills across prospecting, needs assessment, presentation, objection handling, negotiation, and closing; interpersonal and communication skills; problem-solving and strategic thinking; and self-management attributes such as resilience, proactivity, adaptability, and ethical conduct.

Training programs also need valid assessment of competency development. Traditional assessments often focus on isolated knowledge recall, whereas competency requires evidence of knowledge and skill applied in realistic contexts. In DGBL, performance within game scenarios, choices made, strategies employed, and outcomes achieved can provide indicators of targeted competency development (All et al., 2021).

The DGBL intervention targets applied sales development through realistic scenarios and active participation. This competency domain is represented through ADAPTS-SV adaptive selling beliefs, transfer intention, and related post-intervention outcomes. Adaptive selling was selected as the focal sales capability because it is the capability most directly exercised by the intervention's core loop. Each game encounter requires the learner to read a customer situation, select a response from constrained alternatives, and observe the consequence. This is a compressed version of the practice cycle through which adaptive selling is expected to develop (Spiro and Weitz, 1990). ADAPTS-SV was used because it is a validated short-form instrument for adaptive selling beliefs (Robinson et al., 2002), giving the study a defensible measure of applied sales capability without adding excessive survey burden. From this point forward, the thesis uses *performance* as shorthand for adaptive selling performance operationalized through ADAPTS-SV. This is a validated self-report outcome, not an observed workplace sales metric.

2.3 Theoretical Integration

2.3.1 Cognitive Load and Self-Efficacy

Cognitive Load Theory and Self-Efficacy Theory come from different psychological traditions, but both describe processes that interact in complex skill acquisition such as corporate sales training.

Cognitive load during learning can affect self-efficacy. Poorly structured information, confusing interfaces, or too many novel elements can impose extraneous load and reduce the capacity available for learning (Sweller et al., 2011). Such situations can produce frustration, confusion, and a sense of incompetence. These experiences undermine mastery experiences, which Bandura (1977) identifies as the strongest source of self-efficacy. Excessive cognitive load can also induce anxiety or mental fatigue (Chang et al., 2018), which learners may interpret as evidence of inadequacy.

Existing self-efficacy can also shape the perception and management of cognitive load. Learners with stronger self-efficacy tend to approach challenging tasks with greater persistence, effort, and strategic thinking (Bandura, 1986). They may allocate cognitive resources more effectively and experience difficult tasks as engaging. Learners with low self-efficacy may feel overwhelmed sooner and disengage earlier. Without effective cognitive load management, high load can lead to frustration, lower self-efficacy, reduced persistence, and further failure.

Instructional designs that manage cognitive load also create conditions for self-efficacy. When learning tasks feel manageable and achievable, learners can experience incremental success and build confidence through mastery experiences.

The proposed DGBL intervention uses scaffolding and phased increases in task complexity to interrupt cycles of overload and diminished confidence. Its intended pathway is that manageable cognitive demands support mastery experiences, positive affective states, stronger self-efficacy, greater engagement, and more effective application of learned skills.

2.3.2 DGBL as a Cognitive-Cost Tradeoff

A key premise behind using DGBL in complex professional domains such as sales is that interactivity can organize learning in manageable, confidence-building ways. Card-based games can segment decisions, constrain options, and provide immediate contextual feedback, which can help learners cope with complexity (Kordaki and Gousiou, 2017). Kordaki and Gousiou also warn that many digital card games lack explicit scaffolding during play. That expectation and risk informed the design: the card structure was intended to scaffold adaptive selling, and the empirical question is whether that scaffolding was cognitively absorbable for novices.

Every mechanic added to a game introduces its own cognitive cost. Learners must decode the interface, understand the token economy, infer the role of each card type, and coordinate those mechanics with the domain content. For some learners this inventory may become a productive challenge. For learners with weak prior schemas, it may become overhead that competes with domain processing. The design question is whether the intended audience can absorb the cost of an aligned mechanic while still learning from it.

Mayer's (2009) cognitive theory of multimedia learning sharpens the same concern from the multimedia side. The coherence principle suggests that interesting but unnecessary material can depress learning when it consumes processing capacity without contributing to schema construction, and the signalling principle highlights the need to direct attention toward relevant material. Game mechanics occupy an ambiguous position under these principles: a token economy or card rule may be instructionally meaningful in design while still competing for limited processing capacity before learners understand how it serves the sales task.

This study evaluates that tradeoff under controlled conditions because the game and control arms were built around the same sales content. Both conditions covered the same sales stages, used the same scenario family, and included analogous quick-check decisions. The main difference was mechanics inventory. This makes the intervention suited to asking whether a richer mechanics layer improves learning, imposes cognitive cost, or does both depending on learner readiness.

Chapter 3

System Design and Development

This chapter describes the two digital training systems evaluated in the trial and explains the design logic behind the scaffolded sales game. The two conditions were matched on sales content while differing in mechanics inventory. The game was designed through LM–GM alignment, cognitive-load scaffolding, and self-efficacy theory; the empirical test asks whether that mechanics layer added value, cognitive cost, or both.

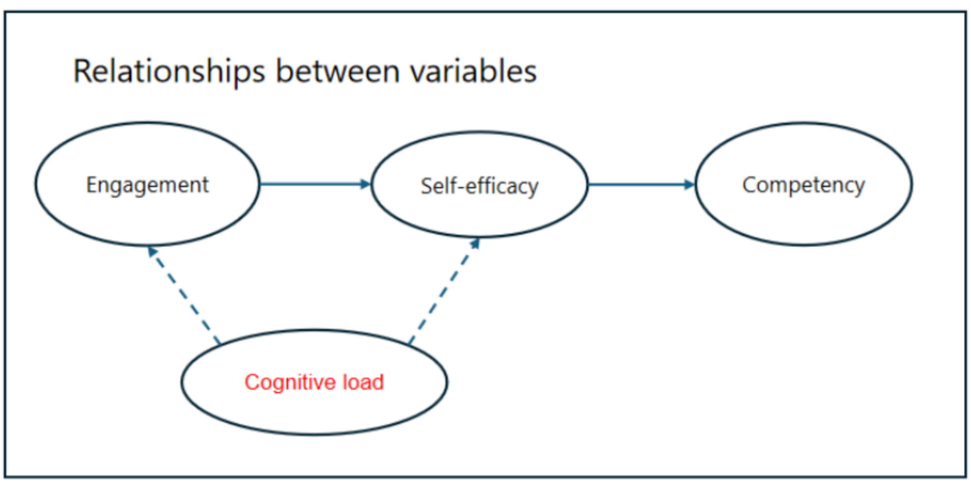

Figure 3.1: Conceptual framework of the DGBL intervention.

3.1 Design Problem

The design problem was to teach adaptive sales reasoning in a short online training environment. The learning content covered recurring sales-process elements, including prospecting, needs analysis, presentation of solutions, handling objections, and closing. In both conditions, the instructional target was not workplace sales performance directly, but the nearer training outcome of performance as operationalized by ADAPTS-SV, together with self-efficacy, engagement, transfer intention, and cognitive load.

The central design assumption was that game mechanics could make sales scenarios more active and confidence-building than a conventional digital module. In LM–GM terms, classification, card choice, token progression, scenario encounters, and feedback were selected because they mapped onto learning mechanics such as identification, repetition, contextual discrimination, application, and reflection. In CLT terms, the game attempted to segment the sales process into smaller decisions and provide feedback before moving learners into more complex encounters. In self-efficacy terms, the game attempted to create mastery experiences through successful classification, progression, and repeated scenario practice.

3.2 Game Condition

The experimental intervention was a scaffolded, card-based digital game titled "Improve Your Sales Game." The game used digital cards to represent customer situations, sales stages, actions, and encounter outcomes. Learners first classified scenario cards into sales stages and earned lead tokens. They then used those tokens in scenario encounters that required selecting appropriate responses to customer needs, objections, or sales opportunities. Feedback was delivered immediately through the game state, token outcomes, and encounter results.

The game carried two simultaneous tasks. First, learners had to process the sales content: recognizing stages, interpreting scenarios, and selecting appropriate sales responses. Second, learners had to operate the game system: understanding card roles, token logic, gated progression, and encounter rules. The original design expectation was that the second task would support the first by making practice more concrete and engaging. The trial showed that this added mechanics layer also had a measurable cognitive cost.

Table 3.1: A priori LM–GM alignment of the scaffolded DGBL intervention

Game mechanic	**Intended learning mechanic**	**Bloom level**	**Primary adaptive-sales target**
Event-card classification	Identification, categorisation, repetition with corrective feedback	Retention / Under-standing	Recognising lead cues and matching customer situations to sales stages
Immediate accuracy feedback	Reflection, reinforcement, error correction	Understanding	Clarifying why a lead or response fits a given sales stage
Lead-token economy	Progression incentive, gated access to later tasks	Retention / Applying	Encouraging accurate foundational decisions before scenario play
Scenario encounters	Application, decision practice, contextual discrimination	Applying	Selecting responses to objections, needs, and solution-fit situations
Action-card choice under constraints	Strategic selection, comparison of alternatives	Applying / Under-standing	Choosing tactics appropriate to customer context
Escalation or difficult-situation cards	Evaluation, strategic judgment, metacognitive reflection	Understanding / Applying	Responding to difficult or unexpected sales situations

3.3 Control Condition

The control intervention was designed as a meaningful conventional comparator. It covered the same broad sales topics in a self-directed digital module with 16 text-based screens, menu-based navigation, static visual support, and scenario-based decision prompts. Learners could move non-linearly, skip familiar material, and revisit earlier sections. The control condition excluded the game's card mechanics, token economy, gated phases, and resource-management structures.

This distinction matters for interpretation. Because the control module was content-matched and still included scenario prompts, any game advantage would need to come from the mechanics layer and its motivational structure. The observed cognitive-load gap is more plausibly attributable to the additional mechanics inventory than to greater intrinsic content difficulty, because the sales topics and scenario family were held broadly constant.

3.4 Design Contrast

Table 3.2: Design contrast between the experimental and control conditions

Design dimension	**Scaffolded DGBL condition**	**Control digital module**
Content scope	Prospecting, needs analysis, presentation, objections, closing	Same broad adaptive-sales topics
Navigation	Mechanic-gated progression across game phases	Menu-based self-navigation across 16 screens
Practice format	Card classification, token use, branching encounters	Text-based explanation plus branching scenario prompts
Feedback style	Immediate in-game accuracy and outcome feedback	Content review through slides and scenario prompts without resource-management mechanics
Learner challenge	Requires decoding rules, tokens, and encounter logic while processing sales content	Requires content selection and scenario judgment without game rules
Primary added demand	Mechanics operation plus content processing	Content processing with self-paced navigation

3.5 Original Design Expectations

The study began from the expectation that the aligned game would produce more favorable learning conditions than the conventional module. Specifically, the game was expected to reduce perceived cognitive load through scaffolding, raise engagement through interactive mechanics, and strengthen self-efficacy through repeated mastery experiences. The conceptual framework predicted that cognitive load, engagement, and self-efficacy would form a coherent pathway into performance and transfer intention.

Based on this framework and the research questions, the following hypotheses are proposed:

- **Hypothesis 1.** Participants in the scaffolded DGBL intervention ("Improve Your Sales Game") will demonstrate more favorable sales training outcomes than participants in the content-matched conventional digital module. This hypothesis is supported if the DGBL group reports higher engagement, higher sales self-efficacy, lower perceived cognitive load, greater performance, and higher transfer intention.
- **Hypothesis 2.** Within the scaffolded DGBL sales-training condition, perceived cognitive load, learner engagement, and sales self-efficacy will be interrelated and will collectively contribute to performance in theoretically consistent ways. This hypothesis is supported by significant intercorrelations in the expected directions between perceived cognitive load, sales self-efficacy, and learner engagement, and by positive associations of self-efficacy and engagement with performance, transfer intention, and related applied sales outcomes.

These hypotheses establish the aggregate and within-game tests. The same conceptual framework implies a readiness-contingent pathway: baseline self-efficacy and baseline performance may condition whether learners can absorb the mechanics layer and benefit from the game. This readiness pattern is evaluated as a continuation of the regression logic used for the pre-post outcomes, using condition $\times$ baseline-readiness interactions and fitted interaction plots to show how the game-control difference changes across learner readiness.

Chapter 4

Methodology

This chapter outlines the research methodology employed to address the research questions posed in the introduction of the study. It details the research design, the characteristics of the study participants and sampling strategy, the instruments used for data collection, the experimental procedure, and the plan for data analysis. The design of the experimental and control interventions is documented in Chapter 3.

4.1 Research Design

This study utilized a quantitative, experimental research design, specifically a pre-test, post-test randomized controlled trial. This design was selected for its recognized strength in establishing cause-and-effect relationships between an intervention and observed outcomes, and for its ability to minimize threats to internal validity (Kirk, 2013; Shadish et al., 2004).

The RCT involved two parallel groups:

- **Experimental Group:** Participants received the scaffolded, game-based digital sales-training intervention, designed as a card-based digital game.
- **Control Group:** Participants received the content-matched conventional digital module, which covered the same broad sales concepts without the card mechanics, token economy, gated phases, or resource-management structures of the experimental condition.

Participants were randomly assigned to either the experimental or control group. Data on key dependent variables, including self-efficacy, perceived cognitive load, learner engagement, transfer intention, and performance, were collected before the training intervention and immediately after its completion.

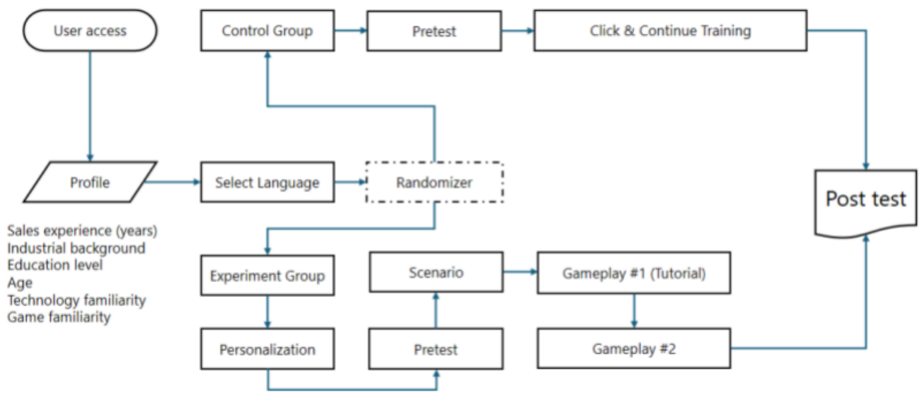

Figure 4.1: RCT design of DGBL in sales education.

The choice of an RCT design is justified by its capacity to address the study's primary research question, which seeks to compare the effectiveness of the two training interventions. Random assignment helps ensure baseline equivalence on known and unknown confounding variables, thereby allowing more confident attribution of observed differences to the intervention itself. The hypotheses and primary outcome comparisons were specified before data collection. Behavioral log data confirm that the first participant session was recorded on 31 March 2026 and the final session on 15 April 2026, establishing the data collection window as a two-week period in which all event-level records in the platform database were generated.

4.2 Population

The target population for this study comprised novice sales professionals or individuals aspiring to enter sales roles who had limited prior formal sales training or experience. Because the study was delivered online through a publicly accessible website, participation was open to any adult individual interested in sales training.

Participants were recruited through online channels, including professional networking sites, relevant online forums or communities focused on sales and professional development, and related digital recruitment routes. Eligibility for inclusion in the final analysis primarily depended on completion of the pre-test, the assigned training module, and the post-test measures.

No direct compensation was planned for participation, but participants benefited from free access to a sales training module. Participation was voluntary, and informed consent was obtained digitally before data collection began.

To characterize the sample, demographic information was collected as part of the pre-test questionnaire. This included:

- Age group (e.g., 18–24, 25–34, 35–44, etc.)
- Years of experience in sales, if any (e.g., none, less than 1 year, 1–3 years, more than 3 years)
- Highest level of education achieved
- Familiarity with digital games for learning
- Current industry of work, if applicable

This demographic data was used descriptively to characterize the sample and to support secondary subgroup interpretation where feasible.

4.3 Measures and Procedure

4.3.1 Instruments and Measures

Data for this study were collected using questionnaires administered at pre-test and post-test, alongside behavioral and applied sales indicators. The measurement design prioritized validated scales where available and theory-informed item development where needed.

- **Sales Self-Efficacy:** Participants' confidence in performing key sales tasks was assessed with a 10-item scale adapted from Bandura's self-efficacy guidelines and prior sales-education work (Bandura, 1977; Knight et al., 2014).

- **Perceived Cognitive Load:** Subjective cognitive load during the training intervention was assessed post-intervention using a 4-item scale informed by Paas et al. (2008), Chang et al. (2020), and Liu et al. (2023).
- **Learner Engagement:** This multifaceted construct was assessed post-intervention using a 6-item scale developed from Fredricks et al. (2004) and related DGBL research (Woo, 2014).
- **Performance (ADAPTS-SV):** Applied sales performance was operationalized through ADAPTS-SV, a validated short-form measure of adaptive selling beliefs. Related post-intervention outcomes captured transfer intention and other psychological responses.
- **Transfer Intention:** Participants' intention to apply learned sales skills in work contexts was assessed post-intervention using a 3-item scale adapted from Tho and Trang (2015).

The study combined these pre-test and post-test self-report instruments with behavioral log data from the training platform. All major scales showed acceptable to excellent internal consistency in the analytic sample.

Table 4.1: Measurement instruments and internal consistency

Scale	α	**Items**	**Rating**
Pre self-efficacy (1–5)	0.93	10	Excellent
Post self-efficacy (1–5)	0.96	10	Excellent
Post cognitive load (1–7)	0.77	4	Acceptable
Post engagement (1–5)	0.88	6	Good
Post transfer intention (1–5)	0.85	3	Good
Pre ADAPTS-SV (1–7)	0.91	5	Excellent
Post ADAPTS-SV (1–7)	0.92	5	Excellent

Self-efficacy was measured with a 10-item sales self-efficacy scale administered at pre-test and post-test on a 5-point agreement scale, drawing on Bandura's self-efficacy guidelines and prior sales-education work (Bandura, 1977; Knight et al., 2014). **Performance** was operationalized using ADAPTS-SV, a validated 5-item psychometric scale assessing adaptive selling beliefs on a 7-point agreement scale (Spiro and Weitz, 1990; Robinson et al., 2002), administered at both pre-test and post-test. The short-form version was selected deliberately: its brevity minimises additional survey burden and reduces the risk of attrition from survey fatigue, which is a particular concern in voluntary online training contexts.

Cognitive load was measured post-test using four items assessing mental effort, perceived difficulty, complexity, and mental strain, following standard subjective load measurement logic (Paas et al., 2003; Paas et al., 2008; Sweller et al., 2011). A validated multidimensional alternative such as Leppink et al.'s (2013) intrinsic/extraneous/germane instrument was not used because the original study instruments had already been finalised for a short online voluntary session, and adding a longer diagnostic battery would have increased survey burden in a design already vulnerable to attrition. **Engagement** was assessed post-test using six items spanning behavioral, emotional, and cognitive engagement, consistent with multidimensional engagement theory (Fredricks et al., 2004; Woo, 2014). **Transfer intention** was assessed post-test using three items measuring the intention to apply learned sales skills in future practice, adapted from transfer-focused workplace learning research (Tho and Trang, 2015).

The survey instrument file indicates a 7-point format for cognitive load and a 5-point format for transfer intention. Scale interpretation and scoring follow those instrument codings.

4.3.2 Procedure

The study was conducted entirely online. The following steps outline the procedure for participant involvement.

Recruitment and Initial Contact

- Participants were recruited through online channels such as professional networking sites, online forums, and related digital communities.
- Recruitment materials provided a brief overview of the study's purpose, the voluntary nature of participation, the estimated time commitment, and the benefit of free access to a sales training module.
- A link to the study website was provided in the recruitment materials.

Informed Consent and Eligibility

- Upon accessing the study website, potential participants were presented with an information sheet explaining the research, data usage, confidentiality, withdrawal rights, and researcher contact details.
- Participants then provided digital informed consent before proceeding.

- A brief screening question was used to ensure that participants were adults.

Pre-Test Data Collection

- After providing consent, participants completed an online pre-test questionnaire.
- The pre-test included demographic information, baseline self-efficacy, and baseline applied sales measures.

Random Assignment

Upon successful completion of the pre-test, participants were randomly assigned to one of the two intervention conditions:

- Experimental Group: Scaffolded DGBL intervention ("Improve Your Sales Game").
- Control Group: Content-matched conventional digital module.

The randomization was managed automatically by the online platform hosting the study using a simple assignment algorithm based on time and sequence of access.

Training Intervention Phase

Participants were then immediately directed to their assigned digital training module.

- The experimental group played the designed DGBL "Improve Your Sales Game."
- The control group completed the content-matched conventional digital module.

Both interventions were designed to cover the same broad sales content and were intended to be completed in a single session.

Post-Test Data Collection

Immediately after completing the assigned intervention, participants completed an online post-test questionnaire. This included post-training self-efficacy, cognitive load, engagement, transfer intention, and adaptive-selling measures.

Participant Flow and Analytic Sample

Participant flow is summarized in a CONSORT-style format to clarify recruitment, randomization, attrition, and the final analytic sample.

The platform recorded 179 individuals who accessed the study website. Of these, 124 completed the demographic form and 92 completed the pre-test and were randomized. Allocation favored the game condition numerically because of the observed platform flow during recruitment: 55 participants were assigned to the game condition and 37 to the control condition. Non-completion was substantially higher in the game condition. Twenty-three of the 55 participants allocated to the game condition did not provide usable post-test data, compared with 6 of the 37 in the control condition. The final per-protocol analytic sample comprised 63 participants: 32 in the game group and 31 in the control group.

This attrition pattern is substantively important. The game group experienced significantly higher attrition than the control group (41.8% vs. 16.2%; Fisher's exact test, odds ratio = 0.28, p = .010). In the game arm, dropout was concentrated during phase 2 of the matching game (41% of game dropouts), phase 1 card classification (24%), and cases in which participants did not engage meaningfully with the game at all (21%). In the control arm, most dropouts occurred during slide training (63% of control dropouts), followed by attrition after the pre-test (25%).

Baseline self-efficacy did not differ significantly between completers and non-completers, suggesting that attrition was not obviously ability-selective. The differential attrition still implies that the final game sample may represent a more persistent or motivated subset of those initially randomized. Accordingly, the primary analyses reported in this thesis are per-protocol analyses based on completers, with the non-completion pattern treated as an outcome-relevant feature of the intervention.

4.4 Data Analysis Plan

Prior to hypothesis testing, data were screened for accuracy, missing values, and the assumptions of statistical tests. Descriptive statistics were calculated for demographic variables and key outcome measures at pre-test and post-test for both groups. An alpha level of .05 was used for tests of statistical significance.

The final completer sample served as the primary analytic set ($N = 63$). For the two primary

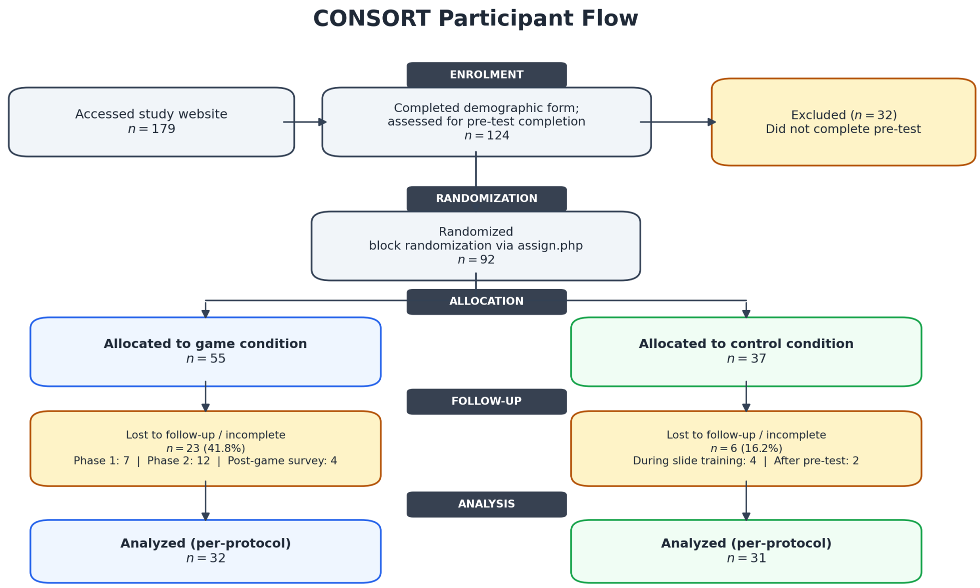

Figure 4.2: CONSORT-style participant flow diagram.

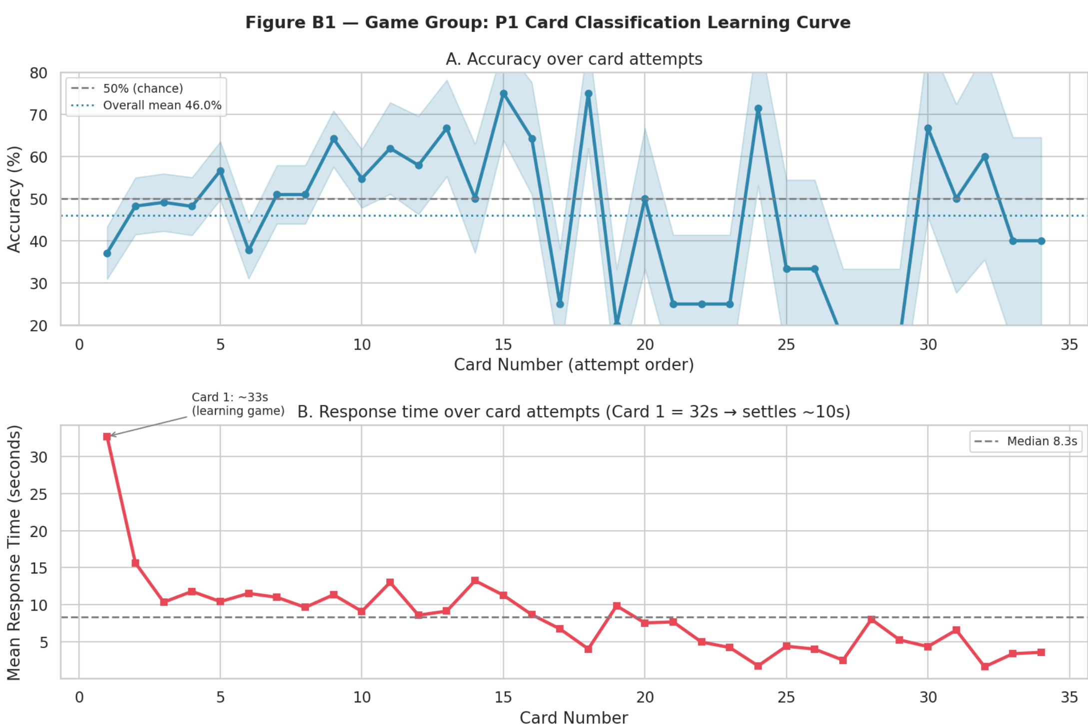

Figure 4.3: Observed attrition points across the game intervention.

Table 4.2: Hypothesis and Analysis Methods

Focus	**Variable(s)**	**Statistical analysis**
RQ1 / H1 between-group comparisons	Self-efficacy, cognitive load, engagement, transfer intention, and performance	Between-group comparisons of post-test outcomes and pre-post change scores across conditions.
Readiness continuation of H1 effects	Condition, pre-test self-efficacy, pre-test performance, post-test self-efficacy, and post-test performance	Regression continuation of the primary ANCOVA logic using condition, centred baseline score, and condition $\times$ baseline interaction terms; fitted interaction plots visualize the simple slopes.
RQ2 / H2 interrelationships within the DGBL group	Perceived cognitive load, learner engagement, sales self-efficacy, and performance	Pearson correlation coefficients, Fisher z comparison of condition-specific correlations, and bootstrapped mediation analysis.
Supporting learning analytics	In-game behavioral telemetry, phase completion, accuracy, response time, and win rate	Descriptive and correlational analysis used to contextualize the psychological results.

pre-post outcomes—sales self-efficacy (SE) and performance—between-group comparisons used ANCOVA with the respective pre-test score as a covariate, following CONSORT recommendations for RCTs that include baseline measurement. ANCOVA was preferred over gain-score t-tests because it does not impose the implicit constraint that the post-on-pre regression slope equals 1.0, and because partitioning pre-test variance from the error term increases statistical precision even when baseline groups are well-balanced. The readiness analysis extended this same regression framework by adding the interaction between condition and the centred baseline score. For SE, the moderator was pre-test SE; for performance, the moderator was pre-test performance. The interaction term tested whether the adjusted game-control difference changed across baseline readiness, and fitted-line plots were used to visualize the simple slopes in the same style as the conceptual framework. For the post-test only measures—cognitive load, engagement, and transfer intention—Welch's independent samples t-tests were used, as no pre-test baseline was available for these constructs. Within-group pre-post changes on self-efficacy and performance were tested using paired t-tests for descriptive completeness only and are not interpreted as causal evidence in isolation.

Exploratory analyses—including correlation comparison via Fisher z tests, three-cluster readiness profiling with k-means, within-game profile translation adapted from Chen et al. (2018), Tukey HSD pairwise contrasts within the game group, and bootstrapped mediation analysis within the game group—were conducted as theoretically motivated follow-ups to the primary hypotheses. For the within-game cluster follow-up, gain scores (post minus pre) were used for pre-post measures (SE change and performance gain), post-test scores were used as-is for post-only measures (cognitive load, engagement, transfer intention), and Tukey HSD post-hoc contrasts were computed on the game group only to identify which expertise-cluster pairs drove any omnibus differentiation. These analyses were not part of the initial hypothesis plan and are reported as exploratory throughout. Sensitivity checks excluded four participants with extreme completion times (more than 10,000 seconds, likely left the browser open); this produced an $n = 59$ sample whose results were consistent in direction and comparable in magnitude to the primary analysis.

Chapter 5

Results and Discussion

5.1 Baseline Equivalence

The two analyzed groups were statistically comparable at baseline. Welch's tests showed no significant differences in pre-test self-efficacy, pre-test performance, or recorded time on task prior to the outcome analyses. This supports the integrity of randomization among completers, while not eliminating the interpretive importance of differential post-randomization attrition.

Table 5.1: Baseline equivalence between analyzed groups (0–100 scale)

Measure	**Game M (SD)**	**Control M (SD)**	t	p
Pre self-efficacy (%)	64.1 (20.0)	62.6 (17.9)	0.33	.746
Pre performance (%)	63.1 (20.4)	62.2 (18.4)	0.20	.851
Time on task (sec)	3592 (9624)	3655 (12333)	-0.02	.982

Note. Self-efficacy and performance pre-test scores are reported on a 0–100 scale for consistency with the ANCOVA analyses in §5.2. Performance refers to adaptive selling performance operationalized through ADAPTS-SV. Both groups are well-matched at baseline. ANCOVA is used regardless of balance; this table describes the starting point.

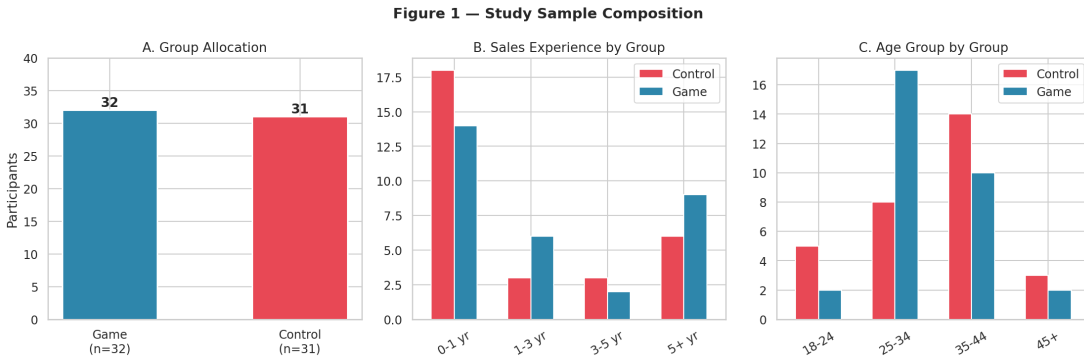

Figure 5.1: Visual comparison of baseline measures between analyzed groups.

5.2 Hypothesis 1: Between-Group Comparisons

Hypothesis 1 predicted that the scaffolded game would produce more favorable post-test outcomes than the conventional digital training format.

Primary Outcomes: ANCOVA (SE and Performance)

For the two primary pre-post outcomes, ANCOVA with pre-test as covariate provided the primary test. Table 5.2 shows results with both unadjusted descriptives and ANCOVA-adjusted means.

Table 5.2: ANCOVA results for primary outcomes (0–100 scale; $n = 63$)

Outcome	**Game M**	**Ctrl M**	**Adj. Game**	**Adj. Ctrl**	F**(1,60)**	p	η_p^2
SE Post (%)	70.2 (17.5)	75.0 (13.5)	69.9	75.5	3.56	.064†	.056
Performance Post (%)	67.9 (19.3)	72.3 (13.7)	67.7	72.5	2.22	.141	.036

Note. Adjusted means estimated at grand mean pre-test score (SE: 63.4; performance: 62.7). Performance is operationalized through ADAPTS-SV. Pre-test covariate $\beta = 0.56$ for both outcomes; model $R^2 = .42$–$.46$. †$p < .10$.

SE showed a marginal trend favouring the control condition ($F(1{,}60) = 3.56$, $p = .064$†, $\eta_p^2 = .056$), approaching a medium effect size. Performance did not differ significantly between groups ($F(1{,}60) = 2.22$, $p = .141$, $\eta_p^2 = .036$, small effect). H1 was not supported for the full sample. The SE effect is still not trivial: at $\eta_p^2 = .056$, the study is underpowered to detect or rule out a group difference of this size reliably. A replication targeting 80% power at $\alpha =$

.05 would require approximately $n = 120$.

ANCOVA interpretation note. These ANCOVA models provide the planned aggregate adjusted comparison for the primary outcomes. The readiness analyses reported in Section 5.6 reveal significant pre-outcome × condition interactions for both SE and performance. The ANCOVA estimates should be read as average adjusted effects across heterogeneous slopes.

Sensitivity Analysis: Intention-to-Treat Estimation

The per-protocol analysis above is restricted to the $N = 63$ participants who completed both the pre-survey and the full training session. To assess robustness to differential attrition (41.8% game vs. 16.2% control), two intention-to-treat (ITT) sensitivity analyses were conducted.

ITT-A: no-gain imputation ($n = 80$). Seventeen non-completers (15 game, 2 control) submitted a pre-survey before discontinuing; their post-test scores were not collected. These participants were included under a conservative no-gain assumption: post-test imputed equal to pre-test (i.e., the training produced zero change for that individual). The remaining 21 non-completers who left before completing the pre-survey were excluded from ITT-A, as no baseline data existed to anchor an imputation.

ITT-B: worst-case bounding ($n = 101$). A broader no-post-test pool identified in the original analysis report was included under extreme assumptions: game non-completers received the lowest observed post-test score on each outcome, control non-completers received the highest. This scenario is an upper-bound exercise and represents the most pessimistic possible estimate for the game condition. It should not be read as a realistic treatment-effect estimate or as the CONSORT completer denominator.

Table 5.3 reports ANCOVA results for SE and performance under all three scenarios.

Note. Adj. Δ = adjusted mean difference (game minus control) in percentage-point terms at grand mean pre-test covariate. PP = per-protocol. ITT-A imputes post = pre for 17 partial completers (pre-survey submitted, training not finished). ITT-B is retained from the original analysis report as a pessimistic bounding scenario over a broader no-post-test pool, not as the CONSORT randomized denominator. $\dagger p < .10$, $*p < .05$, $***p < .001$.

The ITT-A analysis strengthens the SE finding: the marginal per-protocol trend ($p = .063\dagger$) becomes significant ($p = .035^*$) once the 17 partial completers are included under no-gain imputation. The adjusted mean difference in SE is virtually unchanged (-5.6% vs. -5.9%),

Table 5.3: ITT sensitivity analysis: ANCOVA under three attrition scenarios

Scenario	N	**Adj. Δ (G−C)**	F**(1,df)**	p	η_p^2
Self-efficacy (post SE ∼ condition + pre SE)					
PP: completers only	63	−5.6%	$F(1{,}61) = 3.56$	.063†	.056
ITT-A: no-gain ($n = 80$)	80	−5.9%	$F(1{,}78) = 4.69$	.035*	.057
ITT-B: worst-case ($n = 101$)	101	−20.5%	$F(1{,}99) = 31.94$	<.001***	.244
Performance (post performance ∼ condition + pre performance)					
PP: completers only	63	−4.9%	$F(1{,}61) = 2.22$	.139	.036
ITT-A: no-gain ($n = 80$)	80	−2.8%	$F(1{,}78) = 0.91$	.344	.012
ITT-B: worst-case ($n = 101$)	101	−20.6%	$F(1{,}99) = 24.77$	<.001***	.200

indicating that the completer sample attenuated the marginal effect through positive selection. The performance null finding remains non-significant under ITT-A ($p = .344$), confirming that the per-protocol result is not an artefact of survivor selection. Under ITT-B, both outcomes reach significance, but the extreme effect sizes ($\eta_p^2 = .24$ for SE, .20 for performance) reflect the mechanical consequence of assigning all no-data non-completers to floor and ceiling values. Taken together, the ITT analyses suggest that the per-protocol findings are conservative: if anything, the true SE disadvantage for the game condition may be slightly larger than the completer-only analysis suggests.

Secondary Outcomes: Welch t-Tests

Cognitive load, engagement, and transfer intention (post-test only) were compared using Welch's t-tests (Table 5.4).

The only statistically significant between-group difference was cognitive load: the game group reported substantially higher perceived cognitive demand ($t(59.8) = 2.83$, $p = .006$, $d = 0.71$) than the control group—a result opposite to the hypothesized direction. Engagement and transfer intention did not differ significantly, though both favoured the control condition with small-to-medium effects.

Table 5.4: Between-group post-test comparisons: secondary outcomes

Outcome	**Game M (SD)**	**Control M (SD)**	t	p	d
Post cognitive load (1–7)	4.81 (0.83)	4.19 (0.93)	2.83	.006**	0.71
Post engagement (1–5)	3.79 (0.61)	4.03 (0.61)	-1.54	.128	-0.39
Post transfer intention (1–5)	3.80 (0.69)	4.01 (0.62)	-1.27	.209	-0.32

Interpreting the Cognitive Load Gap

The higher cognitive load reported by the game group matters without a post-hoc item decomposition. The two conditions were intentionally matched on domain content: both covered the same sales stages, used the same scenario family, and targeted the same adaptive-selling concepts. Intrinsic load was designed to be broadly constant across conditions. What differed was the mechanics inventory required to access that content. On that basis, the game likely introduced additional non-intrinsic load relative to the control condition. The brief four-item instrument cannot determine whether that additional load was primarily extraneous overhead, germane effort, or a mixture of both.

This measurement limitation should be kept explicit. The cognitive-load scale was selected because the study was conducted in a short voluntary online format where survey burden posed an attrition risk. A richer multidimensional instrument such as Leppink et al.'s (2013) intrinsic/extraneous/germane measure would permit stronger subtype claims in future work. For this study, the conclusion is that the game required more perceived cognitive effort than the control while not delivering superior aggregate performance.

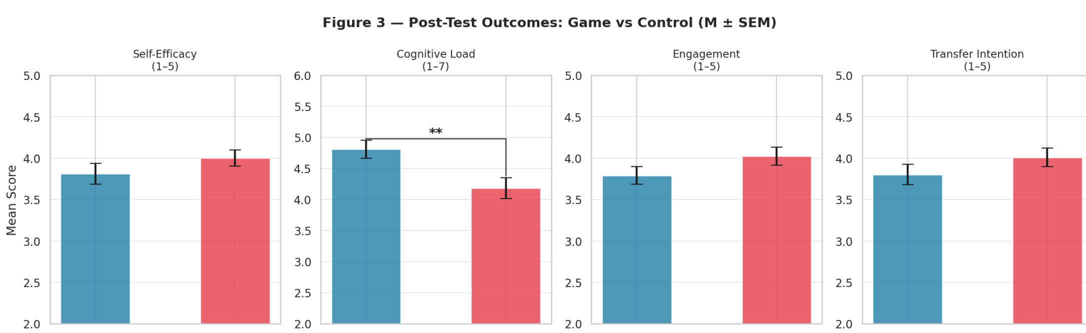

Figure 5.2: Between-group comparison of post-test outcomes and ANCOVA-adjusted means.

5.3 Within-Group Change

Both groups improved significantly from pre-test to post-test on self-efficacy and performance, confirming that both interventions were educationally active. The control group showed larger absolute gains on both outcomes.

Table 5.5: Within-group pre-post improvement summary (0–100 scale)

Group	Measure	Pre M (SD)	Post M (SD)	Gain
Game ($n = 32$)	Self-efficacy (%)	64.1 (20.0)	70.2 (17.5)	+6.2**
Game ($n = 32$)	Performance (%)	63.1 (20.4)	67.9 (19.3)	+4.8
Control ($n = 31$)	Self-efficacy (%)	62.6 (17.9)	75.0 (13.5)	+12.5***
Control ($n = 31$)	Performance (%)	62.2 (18.4)	72.3 (13.7)	+10.1

Note. Within-group SE change: game $p = .003^{**}$; control $p < .001^{***}$. Standardized within-group gains: SE game $d = 0.52$, control $d = 0.74$; performance game $d = 0.42$, control $d = 0.54$.

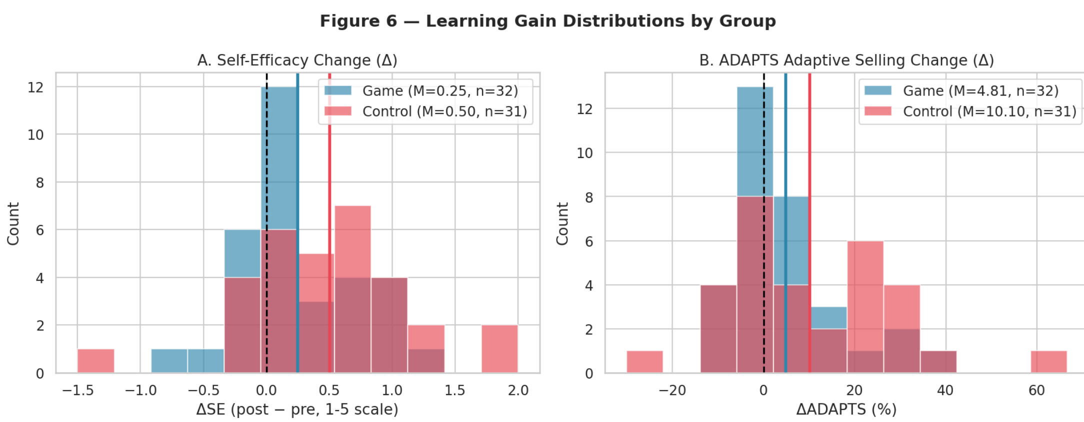

Figure 5.3: Pre- to post-test change in self-efficacy and performance by condition.

Both interventions produced pre-post improvement, with the control group's gains approximately twice as large on self-efficacy. This pattern is consistent with the ANCOVA adjusted means (69.9 vs. 75.5 for SE), where the marginal between-group difference ($p = .064\dagger$) reflects the control group's larger gain, not a starting-point advantage. At the aggregate level, the game was less efficient for this participant mix. The expertise-conditional analysis in Section 5.6 explains why that aggregate result is incomplete.

5.4 Hypothesis 2: Interrelationships Within the Game Group

Hypothesis 2 tested the internal psychological pathway of the game condition. It was examined in three steps: correlations among cognitive load, engagement, self-efficacy, transfer intention, and performance; Fisher z comparisons to check whether those relationships differed from the control condition; and a bootstrapped mediation model testing whether self-efficacy carried the engagement–performance pathway. The results partially supported H2. The positive relationships among self-efficacy, engagement, transfer intention, and performance were strong and coherent. Cognitive load did not show the expected negative relationships with self-efficacy or engagement.

Table 5.6: Selected correlations within the game group

Variable pair	r	p	**Interpretation**
Cognitive load ↔ self-efficacy	.043	.814	Not significant; opposite to expected direction
Cognitive load ↔ engagement	.236	.184	Not significant; opposite to expected direction
Self-efficacy ↔ engagement	.761	$< .001$	Strong positive association
Self-efficacy ↔ transfer intention	.918	$< .001$	Very strong positive association
Engagement ↔ transfer intention	.833	$< .001$	Strong positive association
Self-efficacy ↔ performance post-test	.877	$< .001$	Very strong convergent validity
Engagement ↔ performance post-test	.736	$< .001$	Strong positive association
Cognitive load ↔ performance post-test	.172	.340	Not significant

The strongest message from these correlations is that the motivational and attitudinal constructs formed a tightly connected network inside the game condition. Learners who felt more capable also tended to feel more engaged, more willing to transfer their learning, and stronger on the performance measure. What was not supported was the assumption that lower cognitive load would accompany these benefits. This weakens the argument that the game's scaffolding successfully managed mental effort.

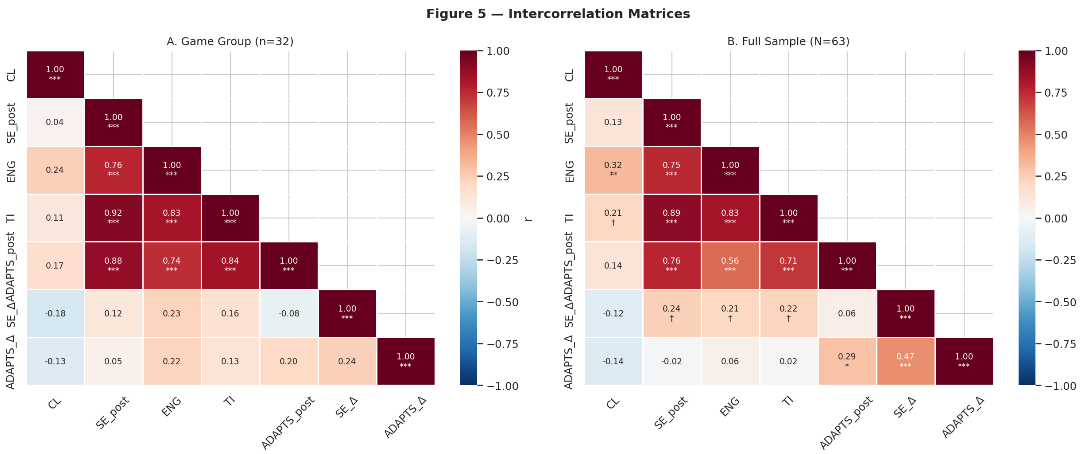

Figure 5.4: Correlation matrix for key psychological and outcome variables within the game group.

Condition Moderation of Construct Relationships (Fisher z)

Fisher z tests compared key bivariate correlations between the game and control groups to assess whether the game restructured the psychological architecture of learning (Table 5.7).

Table 5.7: Condition comparison of construct relationships (Fisher z tests)

Relationship	**Game r**	**Ctrl r**	**Fisher z**	p	**Sig.**
SE × performance	.877***	.532**	2.90	.004	**
ENG × performance	.736***	.305†	2.37	.018	*
CL × ENG	.236 ns	.592***	-1.66	.097	†

The game condition was associated with significantly tighter coupling between self-efficacy and performance ($r = .877$ vs. $.532$, $p = .004$): confidence and adaptive selling beliefs were more tightly integrated in game completers than in control participants. Engagement–performance coupling was similarly amplified ($p = .018$). The strong cognitive load–engagement link present in the control condition ($r = .592$) was attenuated in the game group ($r = .236$, $p = .097$†). This suggests that game mechanics may partially decouple engagement from cognitive effort, consistent with self-determination theory: the game can generate intrinsic engagement through its own mechanics.

Mediation: SE as the Mechanism Linking Engagement to Performance

Bootstrapped mediation analysis (B = 5,000) within the game group tested whether SE mediates the effect of engagement on performance. The indirect effect (ENG $\rightarrow$ SE $\rightarrow$ performance) was significant (indirect effect $ab = 1.10$, 95% CI [.43, 2.24]). When SE was entered as a mediator, the direct effect of engagement on performance dropped from $\beta = 1.41$ to $\beta = 0.31$ (ns), indicating full mediation. SE alone explained $R^2 = .77$ of performance variance ($\beta = .79$, $p < .001$). No cognitive load mediation pathways were significant. Given $n = 32$ and wide confidence intervals, this result is exploratory. Its direction is theoretically coherent: the game's primary within-group learning mechanism runs through confidence-building.

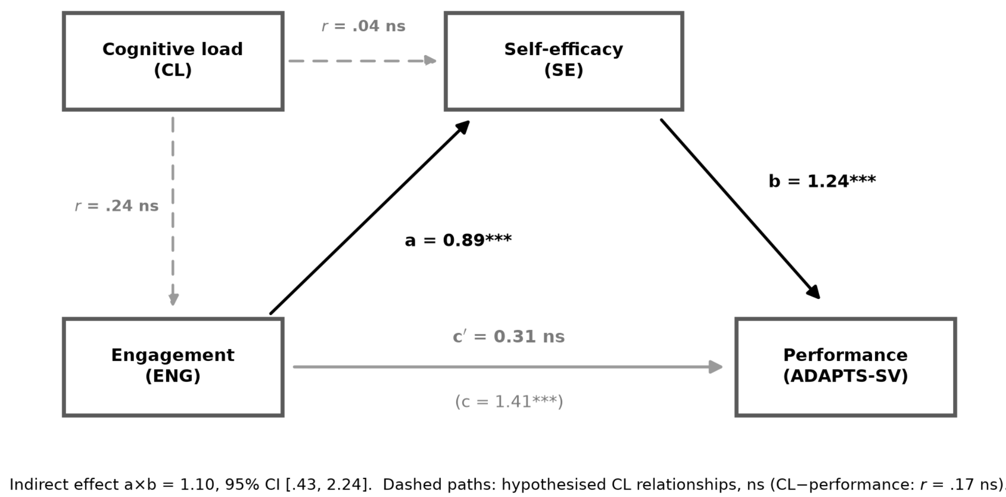

Figure 5.5: Hypothesis 2 path model within the game group ($n = 32$). Solid paths show the self-efficacy mediation pathway from engagement to performance. Dashed paths show the hypothesized cognitive-load relationships, which were not significant. The model supports the motivational pathway while showing that cognitive load did not behave as the original design logic predicted.

5.5 In-Game Behavioral Telemetry

The in-game behavioral telemetry provides context for interpreting why the game may not have produced superior outcomes. In phase 1, average classification accuracy was 57.9%, only modestly above a 50% chance baseline, indicating that the task was genuinely challenging.

Performance varied substantially by sales stage, with follow-up and needs analysis performing relatively well but presentation and pre-approach performing poorly. Faster responders tended to be more accurate and reported higher post-test self-efficacy; the correlation between mean phase-1 response time and post-test self-efficacy was $r = -.37$, $p = .031$. This pattern supports the interpretation that slower, more effortful gameplay was linked to lower confidence.

This use of telemetry is intentionally cautious. Sevcenko et al. (2021) show that theory-grounded in-game metrics can be associated with subjective workload and gaming performance in serious simulation games, and may eventually support real-time adaptation. They also emphasize that behavioral metrics can be influenced by emotion, stress, motor processes, and familiarity with controls. This study uses response time, phase completion, and accuracy as triangulating evidence about gameplay friction, not as standalone cognitive-load measures.

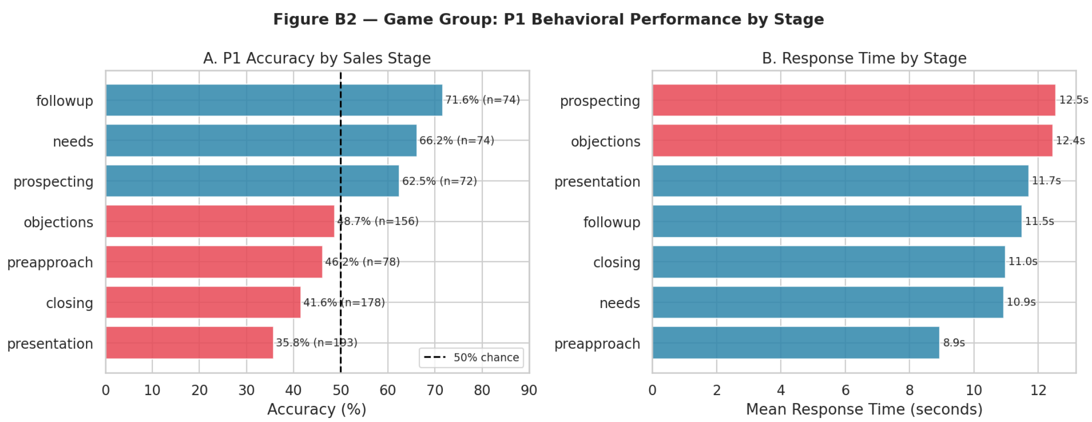

Figure 5.6: Phase 1 classification accuracy by sales stage.

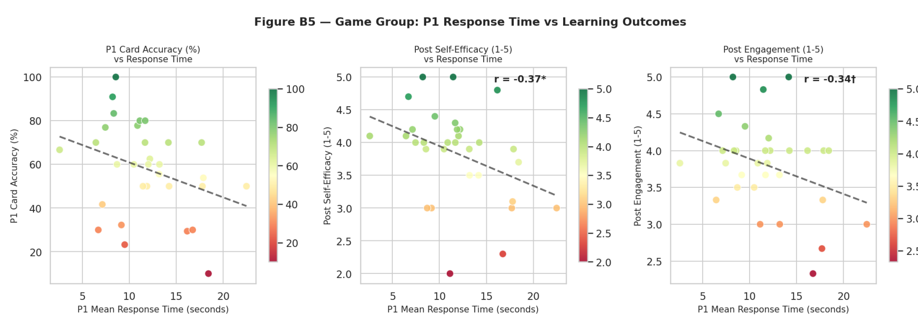

Figure 5.7: Association between phase-1 response time and post-test self-efficacy in the game group.

In phase 2, scenario performance also varied by encounter type. Some event types produced near-perfect win rates, whereas others were much more difficult. Conversion success and win rate correlated positively with engagement, suggesting that behavioral success aligned with a more positive learning experience. Overall gameplay performance did not consistently predict the main learning outcomes strongly enough to treat in-game success as a reliable proxy for learning.

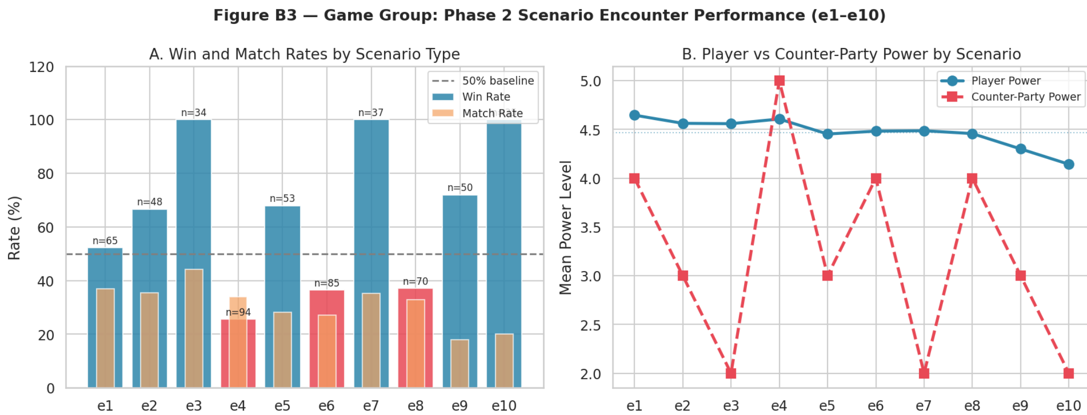

Figure 5.8: Phase 2 win rate by encounter type.

The control module warrants more careful characterisation than a passive "click-through" label suggests. The module was designed with learner agency: participants could navigate content non-linearly via a menu structure, access slides in any order, skip sections they judged to be already understood, and re-visit earlier material. Phase 2 of the control condition consisted of scenario-based decision prompts that paralleled the game's Phase 2 encounter structure, applying the same sales content in a branching-choice format. The control was a scaffolded self-directed module with scenario elements.

Behavioral logs reflect this design. All 31 analyzed control participants viewed all 16 slides; inter-slide timing varied considerably. Twelve participants (39%) progressed with a mean inter-slide gap below 10 seconds, consistent with skipping familiar material or navigating rapidly before returning. Only six participants (19%) averaged more than 30 seconds per slide. This heterogeneity in pacing is consistent with agentic navigation. The control group's equal or better average outcomes came from accessible, self-paced content delivery with genuine interactivity, making it a meaningful comparison to the game.

5.6 Expertise-Conditional Effects

The aggregate between-group comparisons in Section 5.2 obscure substantial heterogeneity across learner expertise levels. To keep the manuscript analytically disciplined, two layers

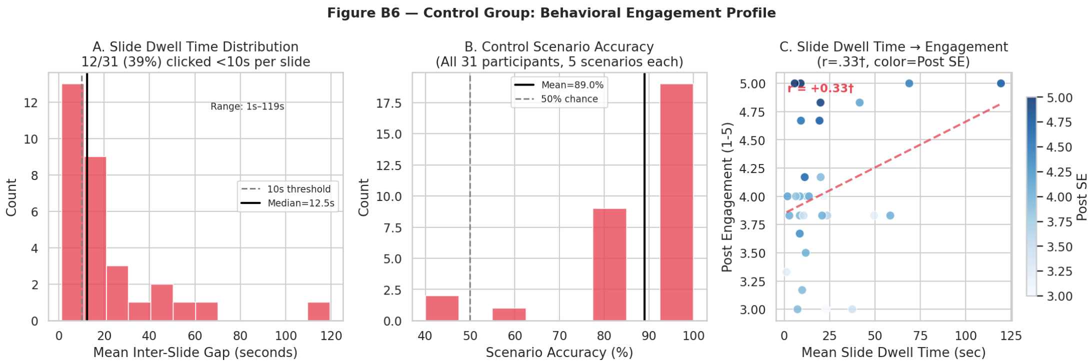

Figure 5.9: Distribution of average inter-slide timing in the control condition.

are distinguished. The *predictive* layer asks *for whom* the game is likely to work and is anchored in baseline readiness variables available before deployment. The *explanatory* layer then asks *what happened inside the game* for those baseline-defined groups, using a within-game profiling table built from psychological outcomes and in-game behavioral telemetry. The predictive layer carries the evidential weight; the explanatory layer makes the pattern easier to interpret.

Readiness Analysis: Simple Slopes and Interaction Plot

OLS regression models of the form *post_outcome* $\sim$ intercept + condition + pre_outcome$_{\text{centred}}$ + condition $\times$ pre_outcome$_{\text{centred}}$ were estimated for each primary outcome. A significant interaction term indicates that the game-versus-control difference depends linearly on pre-intervention expertise level.

Both primary outcomes yielded significant interactions. For performance, the condition $\times$ pre-performance interaction was $F(1,\ 59) = 11.65$, $p = .0012^{**}$ ($\hat{\beta}_{\text{int}} = 0.541$). For self-efficacy, the condition $\times$ pre-SE interaction was $F(1,\ 59) = 5.52$, $p = .022^{*}$ ($\hat{\beta}_{\text{int}} = 0.365$). The simple slopes in Table 5.8 provide the clearest entry point because they translate the interaction into the estimated game-control difference at low, average, and high readiness.

Table 5.8: Simple slopes: game vs. control advantage at three pre-intervention expertise levels

Outcome	**Low (−1 SD)**	**Mean (0 SD)**	**High (+1 SD)**
Performance (pp)	−15.17	−4.83	+5.51
SE (%)	−0.497	−0.224	+0.049

Note. Slope values are the estimated game-minus-control difference in outcome units at each level of pre-intervention expertise. Negative values indicate control advantage; positive values indicate game advantage. Performance is expressed in percentage points; SE in standardised units. Performance is operationalized through ADAPTS-SV.

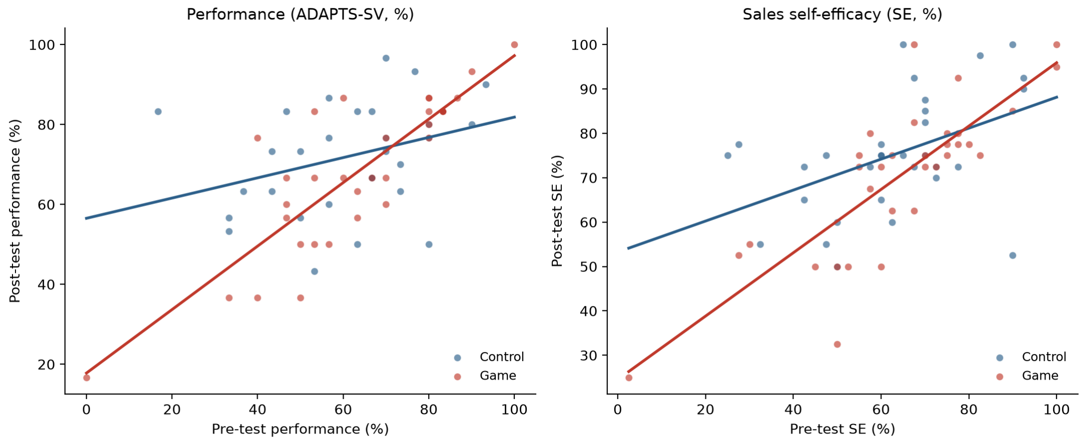

Figure 5.10: Readiness interaction between training condition and pre-intervention scores. Fitted condition-specific regression lines are shown for performance (left panel) and sales self-efficacy (right panel). The steeper game-condition slope visualizes the readiness-conditional pattern summarized in Table 5.8: the game is least favorable at low baseline readiness and becomes increasingly competitive as baseline readiness rises.

The pattern is monotone for both outcomes: the game disadvantages low-readiness participants, produces near-equivalent outcomes at the mean, and trends positively for high-readiness participants. This is the manuscript's primary answer to the thesis question *for whom does the game work?* It is predictive, decision-relevant, and based on variables available before deployment.

Predictive Baseline Clusters (Supporting)

An exploratory k-means cluster analysis (k = 3, silhouette = 0.49) was conducted on two z-standardized *pre-intervention* features only: Pre-SE and pre-performance. Restricting clustering to pre-intervention features avoids circularity: including post-test scores or gain variables in the cluster solution would partially define clusters by the very outcomes used to evaluate them. The k = 3 solution was selected via elbow and silhouette criteria; k = 4 isolated a single outlier and had weaker theoretical value. Silhouette = 0.49 indicates acceptable cluster separation on baseline expertise dimensions. Cluster stability was formally verified by re-running k-means 100 times with different random seeds: the adjusted Rand index (ARI) across all seed pairs was 1.000 and per-participant label agreement was 100%, confirming that the solution is perfectly stable and not an artefact of initialisation.

Table 5.9: Baseline expertise clusters used for readiness profiling ($k = 3$, $n = 63$)

Cluster	n	**G/C**	**Pre-SE**	**Pre-performance**	**Interpretation**
C1 (Novice)	10	4G / 6C	2.31 (low)	33% (low)	Low-readiness entrants
C2 (Intermediate)	28	15G / 13C	3.40	57%	Mid-readiness entrants
C3 (Advanced)	25	13G / 12C	4.17 (high)	81% (high)	High-readiness entrants

Within-stratum game vs. control comparisons on performance gain revealed a monotone readiness gradient across expertise levels, consistent with the learner-tailoring logic of expertise reversal theory (Kalyuga et al., 2003; Kalyuga, 2007):

Table 5.10: Within-stratum game vs. control: performance gain by baseline expertise cluster (small-sample corrected effect sizes)

Expertise stratum	**Game gain**	**Control gain**	**Hedges' g**	**95% CI**	p
Low — C1 (Novice, $n = 10$)	+6.7%	+33.9%	−1.70	[−3.17, −0.23]	.001
Intermediate — C2 ($n = 28$)	+7.6%	+9.7%	−0.15	[−0.89, +0.60]	n.s.
High — C3 (Advanced, $n = 25$)	−1.4%	−1.0%	+0.23	[−0.56, +1.02]	n.s.

Note. Hedges' g applies Hedges' (1981) small-sample correction to Cohen's d, dividing by $J = 1 - 3/(4df - 1)$. 95% CIs computed via the Hedges–Olkin variance formula. C1 raw $d = -1.88$; correction factor $J = 0.903$ ($df = 8$).

Novice learners in the game condition gained substantially less on performance than their control counterparts ($g = -1.70$, 95% CI [−3.17, −0.23], $p < .001$), a large effect consistent with the harm-to-novices side of expertise reversal predictions. The wide confidence interval reflects the small C1 stratum ($n = 10$; $n_{\text{game}} = 4$), and the true effect may be larger due to survivor bias in the game novice group. Intermediate learners showed no clear group difference ($g = -0.15$, CI spanning zero). Advanced learners showed a small, non-significant edge for the game condition on performance ($g = +0.23$, CI spanning zero). The evidence supports a readiness-contingent novice disadvantage more strongly than a full crossover advantage for advanced learners. That asymmetry is plausible in this sample: higher-readiness participants began closer to the upper end of ADAPTS-SV, leaving less room for improvement, and the open online sample may not contain a large enough advanced stratum to estimate the upper side of the reversal precisely. To assess construct validity of the cluster solution, the three clusters were cross-tabulated against three demographic variables not used in the clustering. Sales experience showed a significant, monotone association with cluster membership ($\chi^2(6) = 16.83$, $p = .001$), with mean sales experience increasing from C1 (novice) to C3 (advanced). This confirms that the pre-intervention expertise clusters map onto an independently measured indicator of relevant prior knowledge. Game familiarity ($\chi^2(6) = 1.83$, $p = .886$) and age group ($\chi^2(6) = 9.41$, $p = .162$) showed no significant association with cluster membership, indicating that the clusters are not confounded by gaming experience or age demographics.

These baseline clusters remain exploratory and should not be treated as confirmatory subgroups. Their role here is supportive: they translate the readiness interaction into an interpretable profile and provide the bridge to the within-game explanatory layer below.

Taken together, the aggregate and readiness-stratified results suggest that the game and control conditions followed different instructional pathways. The game imposed higher cognitive cost, and its comparative value depended strongly on learner readiness. The next analyses examine how the internal pattern of experience differed once learners were inside the game, and whether that helps explain why similar completion of the mechanics led to different downstream value across readiness groups.

Within-Game Profile Translation

To make the readiness result more tangible, the baseline clusters were carried into a descriptive analysis of the game arm only. This second layer explains what happened inside the game once learners from different readiness strata were exposed to the same mechanics. It does not answer who should receive the game at entry. Following the table logic used by Chen et al. (2018), psychological outcomes and in-game behavioral telemetry are displayed together, with the outcomes placed first and telemetry treated as supporting evidence. For readability, the three baseline clusters are translated descriptively as a low-engagement novice profile (C1), a moderate intermediate profile (C2), and a highly engaged advanced profile (C3). These are explanatory labels derived from post-treatment patterns, not allocation categories.

The aggregate ordering in the game arm is summarized before the detailed ANOVA table so that the reader can see the profile shape before inspecting pairwise values.

Outcome	**C1**	**C2**	**C3**	
Cognitive load	4.31	4.80	4.98	C3 > C2 > C1
Engagement	3.33	3.65	4.10	C3 > C2 > C1
Transfer intention	3.08	3.56	4.31	C3 > C2 > C1
SE gain	0.78	0.19	0.15	C1 > C2 > C3
Performance gain	6.70	7.57	1.03	C2 > C1 > C3

Table 5.11: Within-game outcome and telemetry patterns by baseline expertise cluster (game group, $n = 32$; adapted from Chen et al., 2018)

Variables	**C1: Novice** *Low-engagement tendency* $n = 4^{\ddagger}$	**C2: Intermediate** *Moderate profile* $n = 15$	**C3: Advanced** *Highly engaged tendency* $n = 13$	F**(2,29)**
SE Gain (Δ, 1–5)	0.78 (0.39)	0.19 (0.45)	0.15 (0.44)	3.311†
Performance Gain (Δ%)	6.70 (8.60)	7.57 (15.09)	1.03 (4.60)	1.245
Cognitive Load (1–7)	4.31 (0.97)	4.80 (0.75)	4.98 (0.88)	1.000
Engagement (1–5)	3.33 (0.24)	3.65 (0.67)	4.10 (0.45)	3.911*
Transfer Intention (1–5)	3.08 (0.42)	3.56 (0.59)	4.31 (0.50)	10.940***
In-game behavioral telemetry				
Phase 1 accuracy (%)	55.00 (20.82)	54.67 (24.46)	59.17 (23.53)	0.128
Phase 2 win rate (%)	55.63 (3.27)	58.07 (12.64)	62.60 (9.28)	0.896
Phase 2 conversions	8.75 (0.50)	8.73 (1.58)	9.25 (0.75)	0.647
Phase 2 failed attempts	7.00 (0.82)	6.47 (2.23)	5.67 (1.67)	0.960
Phase 2 exact matches	5.00 (0.82)	4.80 (2.01)	5.33 (1.72)	0.292
Phase 2 redraws	0.50 (1.00)	1.47 (3.31)	0.00 (0.00)	1.312

Note. Cell entries are M (SD). Performance is operationalized through ADAPTS-SV. ‡C1 novice game-group cell $n = 4$; interpret with caution. Phase 1 accuracy = tokens earned / 10 cards × 100. Phase 2 win rate = conversions / (conversions + failed attempts) × 100. Total time excluded: extreme right-skew due to browser-idle outliers (medians: C1 = 32.8 min, C2 = 17.6 min, C3 = 21.8 min; Kruskal–Wallis $H = 1.46$, $p = .483$, ns). † $p < .10$; * $p < .05$; *** $p < .001$.

The within-game table reveals a dissociation. The psychological outcomes show the clearest profile separation: engagement and transfer intention increased monotonically from C1 to C3, while self-efficacy gain showed a marginal omnibus trend. The in-game behavioral telemetry was more similar across clusters: the groups earned comparable tokens, won comparable proportions of encounters, and differed little on exact matches or failed attempts. The same mechanics inventory could be traversed on roughly similar behavioral footing while producing different subjective and motivational trajectories.

The descriptive labels translate the table into plain language: C1 novices looked like a low-engagement trajectory, C2 like the default moderate trajectory, and C3 like the most engaged trajectory. Those labels are descriptive, not predictive. They show how the predictive readiness model materialised in practice once learners were inside the game. The clearest difference was the quality of the experience attached to performance.

To make the explanatory layer more explicit, Tukey HSD pairwise contrasts were then computed on the game group only. These contrasts do not alter the predictive answer about who should receive the game; they refine the explanatory picture by showing which readiness clusters differed once inside the shared game environment.

Note. Based on observed means. *The mean difference is significant at the .05 level. †$p < .10$. 95% CIs are simultaneous Tukey confidence intervals controlling familywise error ($\alpha = .05$; $df_{error} = 29$, $k = 3$ groups). C1 novice $n = 4$; interpret with caution.

Note. Based on observed means. *The mean difference is significant at the .05 level. †$p < .10$. 95% CIs are simultaneous Tukey confidence intervals controlling familywise error ($\alpha = .05$; $df_{error} = 29$, $k = 3$ groups). Omnibus ANOVA for performance gain and cognitive load were non-significant; Tukey contrasts are shown for completeness because this layer is explanatory. C1 novice $n = 4$; interpret with caution.

The Tukey HSD contrasts sharpen the explanatory picture. Transfer intention is the clearest expertise-differentiated within-game outcome: advanced learners (C3) significantly exceeded both novices (C1; $\Delta = +1.22$, $p = .001$) and intermediate learners (C2; $\Delta = +0.75$, $p = .003$). Engagement shows the same ordering with marginal pairwise contrasts (C3 vs. C1: $p = .055$†; C3 vs. C2: $p = .094$†), consistent with the smaller omnibus effect and the very small C1 cell. For SE gain, no pairwise contrast crosses the .05 threshold despite the marginal omnibus trend, and neither performance gain nor cognitive load differentiates the three clusters within the game arm. This explanatory layer is driven by how motivationally worthwhile the same game felt to different readiness groups.

Table 5.12: Multiple Comparisons (Tukey HSD) — Gain-score outcomes within the game group ($n = 32$)

(I) Cluster	(J) Cluster	Mean Diff (I–J)	Std. Error	Sig.	95% CI Lower	95% CI Upper
Dependent variable: SE Gain (Δ, 1–5 scale)		*$F(2,29) = 3.31$, $p = .051$†, $MSE = 0.194$*				
C1 Novice	C2 Intermediate	+0.5883	0.1751	.061†	−0.0233	1.2000
C1 Novice	C3 Advanced	+0.6212	0.1779	.050†	−0.0003	1.2426
C2 Intermediate	C1 Novice	−0.5883	0.1751	.061†	−1.2000	0.0233
C2 Intermediate	C3 Advanced	+0.0328	0.1179	.979	−0.3790	0.4447
C3 Advanced	C1 Novice	−0.6212	0.1779	.050†	−1.2426	0.0003
C3 Advanced	C2 Intermediate	−0.0328	0.1179	.979	−0.4447	0.3790
Dependent variable: Performance Gain (Δ%, 0–100 scale)			*$F(2,29) = 1.24$, $p = .303$, $MSE = 126.32$ [ns]*			
C1 Novice	C2 Intermediate	−0.8733	4.4722	.990	−16.4929	14.7462
C1 Novice	C3 Advanced	+5.6692	4.5440	.656	−10.2012	21.5397
C2 Intermediate	C1 Novice	+0.8733	4.4722	.990	−14.7462	16.4929
C2 Intermediate	C3 Advanced	+6.5426	3.0115	.289	−3.9753	17.0605
C3 Advanced	C1 Novice	−5.6692	4.5440	.656	−21.5397	10.2012
C3 Advanced	C2 Intermediate	−6.5426	3.0115	.289	−17.0605	3.9753

Table 5.13: Multiple Comparisons (Tukey HSD) — Post-test outcomes within the game group ($n = 32$)

(I) Cluster	(J) Cluster	Mean Diff (I–J)	Std. Error	Sig.	95% CI Lower	95% CI Upper
Dependent variable: Cognitive Load (1–7) $F(2{,}29) = 1.00$, $p = .380$, $MSE = 0.686$ *[ns]*						
C1 Novice	C2 Intermediate	−0.4875	0.3295	.554	−1.6381	0.6631
C1 Novice	C3 Advanced	−0.6683	0.3347	.348	−1.8374	0.5009
C2 Intermediate	C1 Novice	+0.4875	0.3295	.554	−0.6631	1.6381
C2 Intermediate	C3 Advanced	−0.1808	0.2218	.834	−0.9556	0.5941
C3 Advanced	C1 Novice	+0.6683	0.3347	.348	−0.5009	1.8374
C3 Advanced	C2 Intermediate	+0.1808	0.2218	.834	−0.5941	0.9556
Dependent variable: Engagement (1–5) $F(2{,}29) = 3.91$, $p = .031^{*}$, $MSE = 0.308$						
C1 Novice	C2 Intermediate	−0.3128	0.2207	.581	−1.0837	0.4581
C1 Novice	C3 Advanced	−0.7683	0.2243	.055†	−1.5516	0.0150
C2 Intermediate	C1 Novice	+0.3128	0.2207	.581	−0.4581	1.0837
C2 Intermediate	C3 Advanced	−0.4554	0.1486	.094†	−0.9746	0.0637
C3 Advanced	C1 Novice	+0.7683	0.2243	.055†	−0.0150	1.5516
C3 Advanced	C2 Intermediate	+0.4554	0.1486	.094†	−0.0637	0.9746
Dependent variable: Transfer Intention (1–5) $F(2{,}29) = 10.94$, $p < .001$, $MSE = 0.287$						
C1 Novice	C2 Intermediate	−0.4710	0.2133	.278	−1.2160	0.2740
C1 Novice	C3 Advanced	−1.2227*	0.2167	.001	−1.9797	−0.4657
C2 Intermediate	C1 Novice	+0.4710	0.2133	.278	−0.2740	1.2160
C2 Intermediate	C3 Advanced	−0.7517*	0.1436	.003	−1.2534	−0.2500
C3 Advanced	C1 Novice	+1.2227*	0.2167	.001	0.4657	1.9797
C3 Advanced	C2 Intermediate	+0.7517*	0.1436	.003	0.2500	1.2534

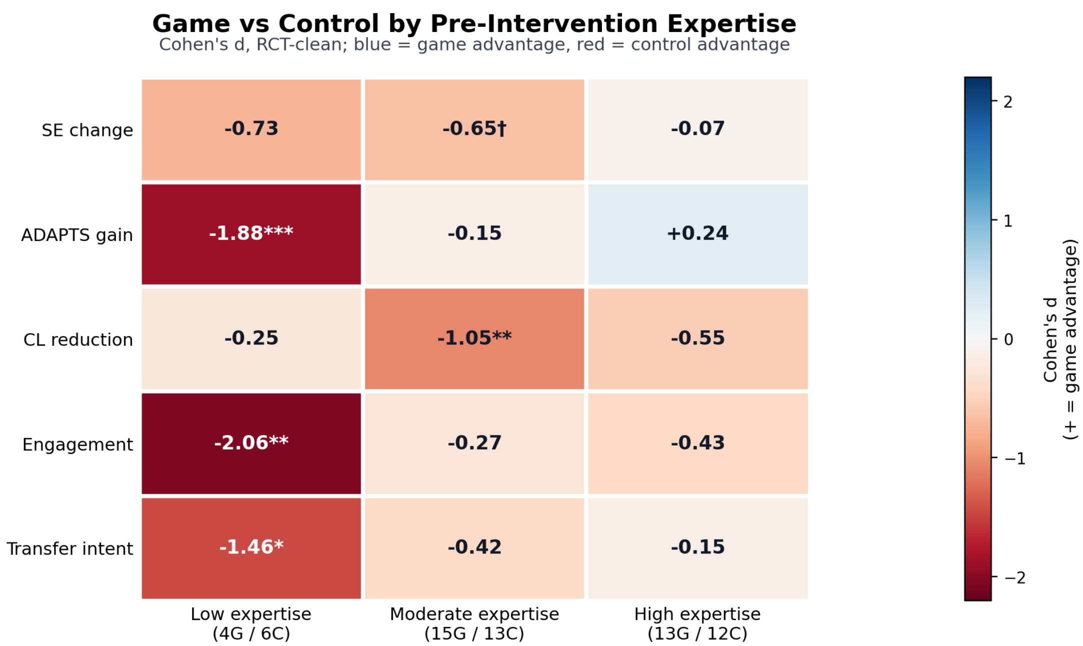

Figure 5.11: Effect sizes (Cohen's d, game vs. control) by outcome and expertise cluster. Values are oriented so positive/blue cells indicate game advantage and negative/red cells indicate control advantage. The cognitive-load row is coded as cognitive-load reduction: positive/blue means the game lowered cognitive load, while negative/red means the game added load. The monotone pattern on performance gain across low, intermediate, and high expertise clusters is consistent with the readiness logic of expertise reversal theory, with strongest evidence for novice harm and weaker evidence for advanced advantage.

5.7 Integrated Discussion

5.7.1 Summary of Findings

This thesis set out to test two linked propositions: first, whether a scaffolded digital game-based intervention would outperform a conventional digital training format in sales training; and second, whether the internal pattern of psychological outcomes inside the game condition would reflect the intended learning logic. Both hypotheses require a qualified answer.

H1 was not supported at the aggregate level. ANCOVA revealed a marginal trend favouring the control group on self-efficacy ($F(1{,}60) = 3.56$, $p = .064$†, $\eta_p^2 = .056$), with no significant difference on performance. The game produced significantly higher cognitive load than the control ($d = 0.71$) without corresponding gains in engagement or transfer intention. The readiness interaction showed that this aggregate comparison averages over a gradient: the game was associated with poorer outcomes at low pre-intervention expertise and increasingly competitive outcomes as pre-intervention expertise increased. The exploratory baseline cluster analysis translated that gradient into concrete learner strata, while the within-game profiling table showed how those strata experienced the same game differently once inside it. The game appears contingent on learner readiness.

This pattern points to a difference in pathway as well as a difference in average magnitude. The control condition appears to preserve a more conventional alignment between accessible instruction, confidence gain, and outcome improvement, particularly for novices. The game condition adds cognitive cost and makes readiness more consequential: advanced learners show the strongest transfer-oriented profile, whereas novice learners can traverse broadly similar mechanics without converting that experience into comparable downstream value. The mediation and correlation analyses test whether the game reorganized the psychological pathway through which learning value was produced.

H2 received partial support. Within the game group, self-efficacy, engagement, transfer intention, and performance formed a tightly coherent positive network, corroborated by the finding that self-efficacy fully mediated engagement effects on performance (indirect effect = 1.10, 95% CI [.43, 2.24]). Cognitive load did not show the expected negative association with other outcomes, suggesting that subjective load captured a more complicated experience than a simple overload story. Fisher z tests further showed that the game significantly tightened the self-efficacy–performance coupling ($r = .877$ vs. $.532$, $p = .004$), indicating that the game restructured the psychological architecture of learning even when aggregate means were not superior.

The high attrition in the game condition (41.8% vs. 16.2% in control) is not merely a methodological footnote; it is part of the substantive evaluation. A format that loses a substantial share of participants before completion — disproportionately around demanding parts of the experience and plausibly over-representing novices — presents real barriers to deployment, irrespective of completer outcomes.

5.7.2 Interpretation Through Cognitive Load Theory

The clearest between-group result was higher cognitive load in the game condition, a finding directly opposed to the intervention's original design intent. The two conditions were intentionally matched on content: both covered the same sales stages, used the same scenario family, and targeted the same adaptive-selling domain. Intrinsic load was designed to be broadly constant across conditions. What differed was the mechanics inventory required to access that content. On that basis, the game likely imposed additional non-intrinsic load relative to the control condition.

The present data do not justify a clean empirical decomposition of that added load into extraneous versus germane components. The brief four-item cognitive-load instrument was chosen for feasibility in a short voluntary online trial, and the earlier post-hoc decomposition attempt does not provide a strong basis for subtype claims. The defensible conclusion is that the game was experienced as cognitively costlier than the control, and that this cost mattered most for learners with the least prior schema to absorb it.

The behavioral data reinforce this reading. Slower processing in the game was associated with lower post-test self-efficacy (r = -.37, p = .031), and phase-1 classification accuracy averaged only 57.9%, only modestly above chance. Many learners appear to have been still mastering the task structure while they were meant to be engaging with sales content. For novice learners specifically, the game concentrated too much difficulty into the early stages before domain schemas existed to absorb it — a pattern consistent with expertise reversal predictions (Kalyuga et al., 2003; Kalyuga, 2007). On theoretical grounds, extraneous overhead is the leading explanation, even if the present instrument cannot isolate it conclusively.

5.7.3 Expertise Reversal Reconsidered

Expertise reversal theory makes a stronger prediction than simple novice difficulty. In its canonical form, instructional supports that help novices can become redundant or inefficient for experts once relevant schemas are automated (Kalyuga et al., 2003). Kalyuga (2007) draws the practical consequence: because the same instructional design can help one learner and harm another, instruction should be adapted to learner expertise, ideally through rapid assessment of what the learner already knows. The present study applies that logic to a mechanics-rich game. The mechanism is comparable: an instructional feature is helpful only when the learner has enough prior schema to absorb its processing demands.

The present evidence is asymmetric and should be stated that way. The novice-harm side is the strongest expertise-related result in the study: the simple slope at low pre-intervention readiness was negative, the low-expertise stratum showed a large control advantage on performance gain, and the attrition pattern is consistent with the same readiness problem. The expert-benefit side is weaker. Advanced learners moved closer to parity and showed the strongest transfer-intention profile inside the game, but their game-control advantage on performance was small and not statistically secure.

Two explanations make this asymmetry plausible without abandoning the expertise-reversal interpretation. First, the advanced cluster entered with high pre-intervention ADAPTS-SV scores, leaving limited headroom for additional immediate gains. A true advantage for advanced learners may be compressed by the bounded scale and immediate post-test timing. Second, expertise was operationalized through baseline self-efficacy and ADAPTS-SV, not through an objective task-based schema test. Some participants classified as advanced may have held strong adaptive-selling beliefs without fully automated domain schemas. The study supports the harm prediction of expertise reversal more strongly than the benefit prediction. A future confirmatory test should combine outcome measures with more headroom, delayed transfer endpoints, and a direct pre-assessment of sales schema automation.

5.7.4 Self-Efficacy, Engagement, and Performance

Within the game group, self-efficacy, engagement, transfer intention, and performance were strongly interrelated, indicating that the motivational side of the theoretical framework was functioning coherently. Learners who felt more capable also felt more engaged, more willing to transfer their learning, and stronger on the performance outcome — consistent with Bandura's (1977, 1986) view that efficacy beliefs are tied to persistence, investment, and anticipated future action.

Mediation analysis showed that self-efficacy *fully* mediated the effect of engagement on performance (indirect effect = 1.10, 95% CI [.43, 2.24]), with the direct engagement–performance path becoming non-significant once self-efficacy was included. The game's primary internal learning mechanism appears to operate through confidence-building: engagement helps when it first builds self-efficacy, which then drives performance. The Fisher z results extend this interpretation: the game significantly tightened the self-efficacy–performance coupling relative to the control condition ($r = .877$ vs. .532, $p = .004$). In the control group, self-efficacy and performance are correlated but more loosely integrated; in the game group they become much more tightly coupled.

The absence of a clear relationship between cognitive load and motivational outcomes (CL $\leftrightarrow$ SE $r = .043$; CL $\leftrightarrow$ ENG $r = .236$, both ns) suggests that subjective load captured a complicated experience. Some learners appear to have experienced the game as effortful but worthwhile, whereas others encountered difficulty without a corresponding gain in confidence. The analyzed game sample, after substantial attrition, was also restricted to more persistent participants, which would attenuate observable load–motivation relationships. The between-group result still stands: the game imposed greater perceived cognitive cost without producing superior aggregate outcomes.

5.7.5 Predictive Readiness and Explanatory Profiles

The analysis distinguishes two legitimate but non-equivalent uses of clustering. The predictive question — *for whom should the game be deployed?* — is answered by baseline readiness and the interaction model. That is the stronger claim because it uses variables available before the intervention. The explanatory question — *what kind of trajectory did different learners experience once inside the game?* — is answered by the within-game profiling table, which translates the statistical pattern into a more tangible narrative of low-engagement novice, moderate intermediate, and highly engaged advanced tendencies.

This separation matters methodologically. Post-treatment profile labels can explain the experience, but they cannot substitute for a predictive allocation rule. Higher-readiness learners are the ones most likely to experience the game as engaging and worth transferring; engagement labels translate that experience after the fact. Keeping the predictive and descriptive layers separate avoids circularity.

The dual-structure analysis also clarifies the role of telemetry. Behavioral logs are not used here as a stand-alone proxy for learning. Instead, they serve as explanatory evidence about how the same mechanics were traversed by learners with different readiness profiles. In that sense, the within-game profiling table is not replacing the predictive model; it is making the predictive model readable.

5.7.6 Cognitive Budget Framework as a Testable Extension

The broader theoretical contribution is best understood by placing LM–GM and the present results inside Plass, Homer and Kinzer's (2015) foundations of game-based learning. LM–GM remains the design-mapping framework: it helps specify which mechanics serve which learning functions. Plass et al. provide the wider anchor by distinguishing cognitive, motivational, affective, and sociocultural foundations. The present study contributes most directly to the *cognitive* foundation by proposing the *Cognitive Budget Framework*: a way to make the cost side of mechanics visible before deployment.

The framework has three components. First, *mechanics acquisition cost* is the working-memory demand created while learners are still learning how to operate the game's interface, rules, card functions, token economy, and feedback logic before those mechanics can carry the intended pedagogy. Second, the *expertise offset* is the relief provided by prior domain schema: a learner who has already automated relevant sales reasoning can spend less working memory interpreting the content itself, leaving more capacity to absorb the mechanics layer. Third, *effective load* is the learner-specific burden that remains after acquisition cost and expertise offset interact. When effective load remains within the learner's workable range, aligned mechanics can become meaningful challenge. When it exceeds that range, aligned mechanics become overhead.

The framework extends LM–GM. LM–GM asks whether a mechanic is aligned to a learning function. The Cognitive Budget Framework adds a learner-readiness question: what will this aligned mechanic cost the learner to acquire, and which learners can absorb that cost? Each component can be operationalized. Mechanics acquisition cost can be estimated at design time by extending the LM–GM table with an acquisition audit: for each mechanic, what must the learner understand before the mechanic teaches anything? It can also be observed empirically through early-phase telemetry such as onboarding time, first-attempt latency, early classification accuracy, and repeated rule-correction events. Expertise offset can be assessed before deployment through self-belief measures such as sales self-efficacy, validated domain measures such as ADAPTS-SV, or more direct cognitive pre-assessments of sales schema. Effective load should then be measured with subjective cognitive-load instruments, ideally multidimensional scales that can distinguish intrinsic, extraneous, and germane load more cleanly than the brief instrument used in this study.

The present study is a worked example of that logic, not a definitive validation. The game and control conditions were content-matched, yet the game produced a substantial cognitive-load gap ($d = 0.71$) and a strong readiness interaction. That pattern is consistent with the Cognitive Budget Framework: the game carried a larger mechanics inventory, and the learners best able to absorb it were those who already possessed stronger prior schema. The framework is defended here as an operational and testable design heuristic, not as a fully validated theory. Its value lies in making explicit something that design frameworks often leave implicit: pedagogical alignment does not guarantee cognitive absorbability.

This reading also clarifies the extension to Arnab et al. (2015). LM–GM can align learning and game mechanics while aligned mechanics still generate a *Bloom-level prerequisite dependency*: lower-level mechanics must first be learned before higher-order learning through those mechanics can begin. For advanced learners, that onboarding cost is small. For novices, it may consume the working memory needed for domain learning. The Cognitive Budget Framework adds a readiness-sensitive cognitive screen to the existing alignment logic. Practically, that screen could guide whether learners receive the full game, a lighter onboarding version, or direct instruction first.

5.7.7 Non-Completers and Deployment

A contribution of this manuscript is to keep non-completers visible instead of collapsing the study into the analyzed sample alone. In online intervention studies, attrition is partly a feature of the setting itself: participants can always opt out. That fact should temper over-interpretation. Attrition patterns still have substantive meaning. In this study, dropout was higher in the game arm, and the behavioral timing data suggest that at least part of that loss was related to friction within the experience.

Attrition has both methodological and substantive roles. Methodologically, it limits the internal security of a pure completer comparison and requires the ITT sensitivity analyses already reported. Substantively, it is consistent with the broader readiness story: a design that asks too much of low-readiness learners will lose some of them before the putative benefits of the game can activate. Non-completion is part of the intervention's real-world evaluation, even though individual dropout reasons cannot be proven from the logs alone.

The practical implication is expertise routing before deployment. The aggregate performance null warns against undifferentiated deployment of game-based adaptive sales training to mixed-readiness audiences in professional-development contexts. For practitioners, the actionable step is pre-training routing using the same baseline survey data already collected in this study. The ROC-derived threshold remains provisional because it was derived from a small exploratory stratum, but the logic is plausible: novices need either direct instruction first or a lighter onboarding layer before the full game becomes appropriate.

5.7.8 Limitations

Several limitations should be acknowledged. First, the final analyzed sample was modest ($N = 63$), and the large post-randomization attrition—especially in the game condition (41.8% vs. 16.2%)—limits generalizability and means the completer comparison is not equivalent to an intention-to-treat estimate. Post-hoc power analysis (Table 5.14) confirms that the study was adequately powered only for the cognitive load comparison; the two primary outcomes were substantially underpowered. The CL comparison (power = 0.792) is likely to replicate; the SE finding (power = 0.462) and performance null (power = 0.313) should be treated as preliminary until replicated in larger samples.

Table 5.14: Retrospective power analysis: required N and achieved power for key outcomes

Outcome	**Obs.** η_p^2 / d	N	α	**Power achieved**	N **for 80% power**
Self-efficacy (SE)	.056	63	.05	0.462	∼140
Performance	.036	63	.05	0.313	∼220
Cognitive load	$d = 0.71$	63	.05	0.792	∼66

Note. Power computed using G^*Power conventions: ANCOVA for SE and performance (covariate ρ estimated from observed pre-post correlation); independent-samples t-test (two-tailed) for CL. $\alpha = .05$ throughout.

Second, the in-game behavioral telemetry adds nuance to the attrition pattern but does not resolve it completely. Without follow-up data from non-completers, the reasons for dropout cannot be confirmed directly. A further interpretive caveat applies to the telemetry more broadly: as Lim et al. (2021) demonstrate in an online learning context, platform trace data capture only the observable portion of a learner's behavior and do not represent the full range of self-regulatory activity that may occur outside the tracked environment. The first-attempt latency and phase-1 accuracy measures used here should be treated as indicators of in-game processing, not comprehensive indices of learning engagement. Examining dropout timing adds nuance: participants who exited within 5 minutes of starting (before completing phase 1 content) are more plausibly responding to cognitive friction and task ambiguity than to time unavailability, since 5 minutes is insufficient for a genuine scheduling conflict to resolve. The MAR assumption is strongest for the 5–15 minute dropout band, where participants had engaged with substantive content before discontinuing; these dropouts are most consistent with a time-tolerance-based decision. Participants in the >15-minute dropout band had completed a substantial proportion of the task, and their non-completion is more puzzling; if their pre-survey data become available through follow-up, inclusion in a prospective ITT-B analysis would be informative.

Third, the exploratory cluster analysis was not part of the initial hypothesis plan. The final cluster solution used two pre-intervention features only (Pre-SE and pre-performance), yielding silhouette = 0.49 — acceptable but not strong separation. This pre-intervention-only approach was adopted to avoid circularity in cluster derivation: an earlier exploratory version of the analysis used 10 features including post-test and gain variables (silhouette = 0.26), which risks partially defining clusters by the very outcomes used to evaluate them; that version was replaced in the final analysis. Stability of the final $k = 3$ solution *was* formally tested across 100 random seeds and yielded ARI = 1.000, indicating that the solution was not an artefact of initialisation. The expertise-reversal finding is asymmetric: evidence for novice harm is stronger than evidence for advanced advantage. The pattern needs confirmation in a fully powered replication. A survivor-bias caveat also applies: game-arm novices who completed the study ($n = 4$) may be the most persistent subset of novice participants, meaning the true game disadvantage for novices could be larger than observed.

Fourth, all primary outcome measures are self-report. The high SE–performance correlation ($r = .877$ in the game group) is partly attributable to construct overlap—both measures tap confidence in sales tasks—and shared method variance from the same post-test session. No objective performance measure was included; the performance findings should be interpreted as self-report beliefs about adaptive selling, not demonstrated skill acquisition.

Fifth, the study focused on short-term post-test outcomes and perceived transfer intention, without long-term follow-up on retention or real-world skill transfer. The transfer intention measure (Tho and Trang, 2015) captures post-training intention, not demonstrated on-the-job transfer. Intention-behavior gaps are well-documented in the training transfer literature (Blume et al., 2010), so transfer intention should be interpreted as a proximal motivational indicator.

Sixth, a format–measurement alignment concern warrants acknowledgement as an alternative explanation for the performance null finding. The control condition delivered content in propositional text slides, and ADAPTS-SV assesses adaptive selling beliefs through propositional, declarative statements about sales behavior (e.g., "I vary my sales style from situation to situation"). The surface format of the measurement instrument more closely matches the control intervention than the game format. If any portion of the performance score reflects familiarity with propositional descriptions of selling concepts, independent of genuine belief change, the control group's advantage may be partially attributable to this format match. This alternative explanation cannot be ruled out from the present data. Future studies comparing game-based and text-based training should consider including scenario-based or process-oriented performance measures that are format-neutral relative to both intervention arms.

Seventh, a measurement timing concern applies to comparative GBL research more broadly, and to this study in particular. Immediate post-test comparisons favour instructional conditions that encode propositions directly: a text-based module that presents propositional content is compared, moments later, on a measure of propositional knowledge or propositional beliefs, producing a near-identical format–timing match. Game-based learning operates through different psychological levers: schema formation through repeated simulation, self-efficacy building through mastery experiences, and motivational priming for continued post-training engagement. These mechanisms are likely to appear more clearly in delayed retention, on-the-job behavioural transfer, and self-directed post-training learning than in immediate post-test measurement. This study partly addressed the problem by using psychometric belief measures (SE and performance) instead of declarative knowledge tests, avoiding the most direct format match. Even belief measures at immediate post-test may capture the control condition's propositional-encoding advantage more directly than the game's motivational–schema pathway. Empirical support for this temporal asymmetry comes from Wouters et al.'s (2013) meta-analysis, which found that the serious game advantage was larger for *retention* ($d = 0.36$) than for *immediate learning* ($d = 0.29$), a pattern consistent with games producing learning through mechanisms that compound after training. The SE-mediated engagement pathway observed within the game group fits this mechanism: the game built confidence and motivational priming that delayed follow-up might capture as continued self-directed learning and behavioural change. Future studies should treat delayed measurement — at a minimum 4–6 weeks post-training, with an on-the-job transfer indicator — as a primary endpoint and frame immediate post-test comparisons as conservative lower-bound estimates of game effectiveness.

Eighth, the sample composition limits generalizability. The study recruited participants from a single professional development context, with approximately 69% working in the education sector and the remainder distributed across industries, predominantly from Southeast Asia, in a voluntary online enrolment setting. This sample profile is unlikely to be representative of sales professionals in other sectors, regions, or organizational training contexts. The expertise distribution (with a relatively small novice stratum, $n = 10$) may also differ from the target population in deployed training contexts, where novices are often the primary intended audience.

Finally, the study was conducted in a web-based self-enrolment environment, which improves ecological realism for digital training but reduces control over attention and context. The control module included genuine learner agency: non-linear menu navigation, content skipping for familiar material, and a Phase 2 scenario component structurally parallel to the game's Phase 2. The heterogeneous pacing observed in control telemetry (39% of control participants with sub-10-second inter-slide gaps) is consistent with agentic selective review and should not be interpreted as evidence that the control was a minimal-effort baseline. Future studies could add explicit measures of within-module navigation behaviour to characterise control condition engagement more precisely.

5.7.9 Future Research

The next step is a fully powered confirmatory replication that stratifies participants by prior sales knowledge before randomisation. A sufficiently powered study (approximately $n = 120$ for 80% power at $\eta_p^2 = .056$) with the readiness interaction specified in advance—using a planned pre-performance $\times$ condition model—would provide stronger evidence about for whom game-based sales training works and under what conditions.

Future designs should also test redesigned game versions that separate interface onboarding from instructional content, with a dedicated non-scored tutorial phase before domain-relevant gameplay begins. Dynamic difficulty adjustment mechanisms—triggered by phase-1 accuracy or response time thresholds—could be tested as a way to route novice learners toward more supported instruction without abandoning the game format entirely (Serrano-Laguna et al., 2012; Sevcenko et al., 2021; Zohaib, 2018). A direct empirical test of the Cognitive Budget Framework would vary mechanics inventory while holding sales content constant, vary learner schema offset through onboarding or prior-knowledge stratification while holding mechanics constant, and replicate the pattern in domains with different procedural demands.

Longer-term follow-up is needed to determine whether the expertise-conditional pattern persists, attenuates, or reverses over time. Delayed measurement should be treated as a primary endpoint in future GBL comparative trials. Immediate post-test comparisons are structurally conservative for game conditions: games operate through schema-building, confidence priming, and motivational carry-forward mechanisms whose outputs may compound after training. A 4–6 week follow-up with behavioural transfer indicators and retention measures would test whether the game's motivational architecture translates into durable learning advantage and would test the delayed-effect prediction implied by the SE-mediation finding. Future studies should also adopt validated multidimensional cognitive load instruments (e.g., Leppink et al.'s 2013 intrinsic/extraneous/germane subscale) to identify which load components drive expertise-conditional effects.

Chapter 6

Conclusion

This thesis evaluated a scaffolded digital game-based intervention for adaptive sales training in a professional-development context against a conventional digital module in an online randomised controlled trial. At the aggregate level, the game did not produce superior outcomes on self-efficacy or performance, and it imposed substantially higher cognitive load than the control ($d = 0.71$). The readiness interaction gives the result its shape: the game was associated with worse outcomes for low-readiness learners and increasingly competitive outcomes as readiness increased.

That predictive readiness pattern is the thesis's main answer to the question *for whom does game-based sales training work?* Engagement is visible only after learners have entered the game, so it cannot serve as the predictive answer. The game works most plausibly for learners who arrive with enough prior schema to absorb the mechanics inventory without exhausting the working memory needed for domain learning. The baseline clusters and the within-game profiling table then make that result readable: novices tended to experience the same mechanics as a lower-value trajectory, while advanced learners were more likely to experience them as engaging and transferable.

The clearest empirical signal enabling this interpretation is the cognitive-load gap. Because the game and control conditions were intentionally matched on sales content, the higher perceived load in the game condition is most plausibly attributable to the additional mechanics layer instead of higher intrinsic content complexity. The present instrument does not allow a clean empirical separation of extraneous and germane load, so that subtype claim remains open. The data show that the cognitive-cost side of game design matters and interacts with learner readiness.

The theoretical contribution follows from this pattern. Arnab et al.'s LM–GM framework helps designers ask whether a game mechanic is aligned to a learning function. Plass, Homer and Kinzer's (2015) foundations of game-based learning show that alignment is only one part of the story; games also rest on cognitive, motivational, affective, and sociocultural foundations. The present thesis extends the *cognitive* foundation by proposing the Cognitive Budget Framework: a design heuristic in which mechanics acquisition cost creates an operating demand, prior schema supplies a learner-specific offset, and remaining working-memory capacity determines whether aligned mechanics function as productive challenge or as overhead. The study is a worked example of that logic, not its final validation. Its value lies in turning a familiar design intuition into a testable proposition.

The motivational findings fit that account. Within the game group, self-efficacy fully mediated the effect of engagement on performance, and the game significantly tightened the self-efficacy–performance relationship relative to the control condition. The game contained a viable internal learning mechanism, and that mechanism appears to have activated mainly among learners who persisted long enough and entered with enough readiness to benefit from it.

Practically, the study supports selective deployment for digital game-based sales training. Game-based training should be deployed with readiness checks for mixed-readiness audiences. Expertise routing can be based on short self-belief instruments, validated domain-readiness measures, or cognitive pre-assessments that approximate whether learners have enough schema offset to absorb the mechanics acquisition cost. Lighter onboarding for novices and dynamic difficulty adjustment become design necessities when mechanics cost is high. The attrition pattern strengthens that conclusion: even in a voluntary online context where some dropout is unavoidable, a format that disproportionately loses lower-readiness learners before completion must count that loss as part of its intervention effect.

The study ends with a narrower and stronger claim than the one it began with. Scaffolded game-based sales training is a precision intervention whose value depends on whether the audience can absorb the mechanics inventory that carries the pedagogy. That conditionality is the central contribution of the thesis.

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